

MotorX

Second-Hand Vehicle Marketplace with Intelligent Search and Automated Inventory Processing

Master Test Plan and Test Evaluation Report

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Item	Detail
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Revision History

Date	Version	Description	Author
21 Sep 2026	1.0	First MotorX plan on the supplied template: target items, eight technique tables, 48 cases and an automated baseline of 77 passing tests.	Group 23
26 Sep 2026	2.0	All suites re-run on Node 24 against a real MongoDB replica set, Redis and MinIO. HTTP journeys, frontend component tests, the 5,000-row benchmark, the crash drill and a live smoke test added; new features covered; the merge with branch version4 retested.	Group 23
27 Sep 2026	3.0	Reorganised into test levels, test types and a level × type matrix, with a case inventory and sections on environments, criteria, schedule, defects and test data, plus appendices A–G. EC2 deployment tested (E2E-A3, CT-04, CT-05); deployment findings F-11 to F-14 added; contract and consistency checks CT-01 to CT-07 introduced. Code coverage measured; screenshot evidence from the EC2 deployment added (Appendix H) with findings F-15 to F-17.	Group 23
28 Sep 2026	3.1	Tool outputs added to Section 5.1: test logs (5.1.1), user interface test details (5.1.2), type-check and dependency audit (5.1.3) browser performance (5.1.4) server monitoring (5.1.5) and database performance (5.1.6); finding F-18 recorded; the 5.1 assessment now counts plain HTTP on EC2 (F-15) against acceptance; figures numbered automatically; frontend component tests UT-42–UT-48 added (frontend coverage 38% → 69%). R-05 closed after the production build.	Group 23

Table of Contents

Revision History	2
Table of Contents	3
1. Evaluation Mission and Test Motivation	5
1.1 What this cycle must demonstrate	5
2. Target Test Items	6
2.1 Items under test	6
2.2 Items outside this cycle	6
3. Test Approach	7
3.1 How the approach is organised	7
3.2 Test Levels	7
3.3 Testing Techniques and Types	9
3.4 Level × Technique Matrix	12
4. Test Case Inventory	14
4.1 Identity and access	14
4.2 Dealer onboarding and profiles	14
4.3 Listings and photos	14
4.4 Buyer discovery	15
4.5 Inventory import and the worker	15
4.6 Dealer stock tools	15
4.7 Administration and notifications	16
4.8 Phones, languages and accessibility	16
4.9 Shared contracts, configuration and deployment	16
5. Deliverables	17
5.1 Test Evaluation Summaries	17
5.2 Reporting on Test Coverage	26
6. Risks, Dependencies, Assumptions, and Constraints	28
7. References	29
8. Responsibilities, Staffing and Training Needs	30
8.1 Ownership by Test Type	30
8.2 Shared Duties	30
8.3 Skills to Build	30
8.4 Staffing Limits	31
9. Test Environment Specification	32
9.1 Differences Between Environments	32
10. Entry, Exit and Suspension Criteria	33

10.1 Entry Criteria	33
10.2 Exit Criteria	33
10.3 Submission Exit Criteria	33
10.4 Suspension and Resumption	33
11. Test Schedule and Phasing	35
11.1 Activities Across Phases	35
12. Defect Severity and Priority Definitions	36
12.1 Severity	36
12.2 Priority	36
12.3 Recording and Lifecycle	36
13. Test Data Policy	37
13.1 Rules	37
13.2 Test Data in Each Environment	37
13.3 Invalid and Hostile Inputs	37
Appendix A – Unit and Component Test Cases	38
Appendix B – Integration and Contract Test Cases	50
Appendix C – End-to-End Test Cases	56
Appendix D – Non-functional Cases and Execution Results	61
D.1 Results by test file	62
Appendix E – Evidence Index	64
Appendix F – Defect Log	65
Appendix G – Traceability Matrix	67
Appendix H – Screenshot Evidence	68

1. Evaluation Mission and Test Motivation

MotorX is Group 23's marketplace for used vehicles in Sri Lanka. Buyers browse and search without an account, dealers must be vetted by an administrator before they can list stock, and administrators approve dealers, moderate listings and watch the system. The code is a TypeScript monorepo with four npm workspaces: a React single-page application (in English, Sinhala and Tamil), an Express API organised as feature modules, a separate BullMQ worker for heavy background jobs, and a shared-contracts package of Zod schemas and types that the other three import. MongoDB Atlas holds the records; Redis carries the job queue and the rate-limit counters; photos, CSV files and dealer documents are kept in S3-compatible storage (MinIO in the deployed stack); Firebase provides sign-in; a Hugging Face sentence-embedding model with Atlas Vector Search supports search by meaning; and e-mail leaves through SMTP via an outbox. For this cycle the system runs on a single Amazon EC2 instance with Docker Compose [1]–[3].

Most of the catalogue never passes through a person's hands. A dealer submits one CSV file, and optionally a ZIP of photos, and the worker validates, normalises, de-duplicates, embeds and stores thousands of vehicles by itself. When that goes wrong it goes wrong quietly — a car listed twice, a photo on the wrong car, a row that simply disappears — and the first person to notice may be a buyer phoning about a car that is not there. A second pressure is privacy: many competing dealers share one database and one API, so every dealer's drafts, uploads, documents and notifications must stay out of every other dealer's reach.

This cycle therefore has three aims. First, to replace claims with evidence: every quality statement in this report points to an executed test, an evidence file or a numbered finding. Second, to find the defects that would do the most damage at submission and in a live demonstration — data leaking between dealers, lost or duplicated stock, jobs that do not recover after a crash, a deployment that exposes internal services, and pages that break on phones. Third, to keep “planned”, “executed” and “passed” clearly separate, so that the team, the mentor and the examiners can see exactly what has and has not been shown.

Position at the time of writing: 348 automated tests executed and all passed (backend 147, worker 94, frontend 50); three system-level procedures executed, including a smoke run against the EC2 deployment; five of seven contract and consistency checks passed and one failed; 0 of the 15 browser journeys executed. Fourteen findings are recorded in Appendix F.

1.1 What this cycle must demonstrate

- A vehicle obeys the same category and powertrain rules whichever way it enters: the listing form, a CSV row or a bulk action (petrol and diesel need an engine size; electric needs battery data).
- A registration number can be open — draft or active — only once, including when two requests race and when an import is repeated.
- A worker that is killed during an import is replaced and the job finishes with exactly one listing per CSV row.
- No dealer can read, edit, publish, re-price or delete another dealer's listings, uploads, documents or notifications by trying their IDs.
- A suspended dealer's vehicles disappear from browse, search and details and come back when the account is reinstated.
- Photos and documents are rebuilt from their pixels or checked by content before anyone can download them: no GPS location, no hidden payload, no disguised program.
- Importing about 5,000 vehicles meets PSR-05, and ordinary API calls on the deployed stack stay well inside PSR-01.
- The buyer site can be used on a phone and in Sinhala and Tamil; compare, similar vehicles, recommendations and the dealer stock tools behave as specified.
- The deployed EC2 stack answers its health checks and exposes only the website and the API.
- Every result can be traced to an SRS requirement, or is clearly marked as an extension beyond the SRS.

2. Target Test Items

The items below are what this cycle tests. Paths are relative to the repository root, so each item can be found in the code.

2.1 Items under test

Test item	Where it lives	Behaviour under test
Identity and access	apps/backend/src/modules/auth-users; shared/middleware; apps/frontend/src/features/auth	Firebase token check with a 2-minute cache and revocation; local user record; role guards; suspension; rate limits; clearing cached data when the signed-in account changes.
Dealer onboarding	modules/dealers; modules/admin (review)	Applications with content-checked documents; verified e-mail required before approval; rejection with a reason and resubmission; profile edits; documents deleted 90 days after a decision.
Listings and photos	modules/marketplace; scripts/migrateListingImages.ts	Category-specific validation; status lifecycle; one open listing per plate; photos re-encoded with an 800 px copy; the public photo route; the photo migration.
Buyer discovery	modules/buyers; modules/search; frontend buyer portal	Filters; natural-language parsing and typo correction; lexical and vector ranking with fallback; similar vehicles; recommendations from recently viewed IDs; side-by-side comparison.
Inventory import	modules/inventory; apps/worker (pipeline, services, jobs)	Upload acceptance, including while Redis is down; queueing; CSV extraction, normalisation, validation and transformation; duplicates; rejected rows; leases and checkpoints; ZIP photo matching; retries; the reaper for lost jobs.
Dealer stock tools (extension)	marketplace listing service; frontend ListingManager and UploadDetails	Bulk publish, mark sold, archive, confirm, re-price and delete; publish every draft of one upload; stale-stock counts and the weekly reminder.
Administration and notifications	modules/admin; modules/notifications; worker e-mail outbox	Dashboard and deep links; upload and audit filters; suspension; audited document views; in-app inbox; e-mail retries.
Frontend experience (extension)	apps/frontend shared components and portals	Drawer menu, filter sheet, swipeable gallery, contact bar with WhatsApp, tables shown as cards on phones, Sinhala and Tamil text.
Shared contracts	packages/shared-contracts	Zod schemas, DTO types, queue job options and IDs, image limits — used by all three applications.
Runtime and delivery	infrastructure/docker; compose.yml, compose.test.yml, compose.ec2.yml; .github/workflows	Node 24 images; the isolated test stack; the EC2 override; CI build, tests and security scans; health endpoints.

2.2 Items outside this cycle

- Payments, finance, chat and saved searches — not part of the implemented system.
- The internal behaviour of Firebase, MongoDB Atlas, Hugging Face and AWS themselves.
- The Amazon ECS deployment path defined in .github/workflows/deploy.yml; the live system for this cycle is a single EC2 instance.
- RR-01 (99% monthly availability), which short test runs cannot show and which needs monitoring over time.

3. Test Approach

MotorX combines three kinds of software with very different ways of failing: request-and-response code in the API, long-running background jobs in the worker, and a browser application used on phones as well as desktops. A single testing style would suit one of these and miss problems in the others, so the approach mixes five kinds of work: quick isolated tests for rules, real-database tests for anything the database enforces, HTTP journeys for authorisation and complete use cases, drills and smoke runs on running containers for recovery and deployment, and browser sessions performed by people for what users actually see. Every finding that has been fixed is now guarded by an automated test or a scripted check, listed against it in Appendix F.

The defects the approach is designed to catch fall into seven groups: rule defects (a vehicle wrongly accepted or refused); identity defects (two open listings for one plate, one CSV row imported twice); interruption defects (a lost lease, a lost queue message, work done twice after a restart); isolation defects (one dealer reaching another's data, a suspended account still visible); content-safety defects (location data or code hidden in an upload); configuration defects (an internal port exposed, a host name refused, a stored address that no longer resolves); and presentation defects (pages that do not fit a phone, text left untranslated).

3.1 How the approach is organised

Each test is described by two independent properties: where it runs, and what quality it provides evidence for. The first gives the four test levels in Section 3.2, from one function in isolation to a person using the deployed site. The second gives the eight test types in Section 3.3, which follow the supplied template; for each type this report also names the ISO/IEC 25010 product-quality characteristic it supports — functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability or portability — so the evidence can be read against a published quality model [4].

The two properties are kept apart because in MotorX they genuinely cut across each other. Dealer isolation, a security property, is checked by a React route-guard test, by HTTP journeys that swap one dealer's IDs for another's, and by browser journey E2E-11: one quality, three levels. In the other direction, a single HTTP journey file makes security, functional and data-integrity assertions in the same run: one level, three qualities. Section 3.4 shows, for every pairing, which cases actually provide the evidence.

3.2 Test Levels

Four levels are used. For each, the table states its scope in MotorX, the tools, what it achieved in this cycle and what it leaves to the other levels.

3.2.1 Unit and Component Testing

At this level one function, schema, service or React component is tested with everything outside the process replaced — MongoDB, Redis, object storage, Firebase, SMTP and the network. The worker's import and photo services are also tested here with mocked repositories; that is what makes it possible to “crash” between two batches or fail the third photo upload on demand, which is hard to arrange with real services.

Aspect	MotorX
Scope	Backend validation schemas, similarity scoring, image and document sanitisation, the token cache, queue publishing and the photo migration (with an in-memory bucket); worker pipeline stages, retry classification, lease renewal, the reaper, e-mail outbox, document retention, stale-stock reminders, and the import and photo services; frontend pages and components rendered in jsdom with the API replaced.
Tools	Vitest 4 (backend, worker) and Vitest 5 with jsdom (frontend); Testing Library and user-event; vi.mock at process boundaries; sharp to generate test images.
Evidence this cycle	291 tests in 48 files; all passed (Appendix A).
Left to other levels	Real queries and indexes; the order in which Express middleware runs; CSS layout; timing between real processes.

3.2.2 Integration Testing

Integration tests cross a real boundary. For MotorX the most important one is MongoDB: unique partial indexes, transactions and atomic updates are enforced by the database, so a mocked repository cannot prove them. These tests run against a MongoDB 7 replica set — transactions need one — in a separate Compose project, motorx-test, which shares nothing with the development stack. A second group, the HTTP journeys, starts the real Express application and sends requests through every middleware to the real database; only Firebase token verification and the S3 client are replaced, because both are external services that cannot be created fresh for each test.

Aspect	MotorX
Scope	Repositories (applications, pending order, registration identity), listing updates, audited document access, rate limits on real Redis, lease ownership and idempotent writes, photo URL updates on keys containing dots; HTTP journeys for access control, the dealer lifecycle, stock tools and discovery.
Tools	Vitest, Mongoose, supertest; compose.test.yml (mongo:7 with replica set rs0, redis:7, MinIO); a guard that accepts only a motorx_test database on a local host; one test file at a time (--maxWorkers=1).
Evidence this cycle	56 tests: 24 data-integration and 32 HTTP-journey; all passed (Appendix B). IT-12 to IT-15 — real Firebase, real SMTP, the queue with a live worker, and a relevance dataset — are planned.
Left to other levels	Real Firebase and SMTP behaviour, browser rendering, and behaviour under many simultaneous users.

3.2.3 Contract and Consistency Testing

MotorX has no network contracts between separately deployed services; its contracts are shared definitions that several parts of the code must agree on. The backend, the worker and the frontend all import packages/shared-contracts; the backend and the worker each declare their own Mongoose models for the same collections; the Sinhala and Tamil dictionaries must match the English one; the API and the worker's reaper must use the same BullMQ job IDs; and the Compose files must publish exactly the intended ports. A change on one side can pass every test around it and still break the other side, so these agreements get their own checks (CT-01 to CT-07).

Aspect	MotorX
Scope	Shared types compiled into every consumer; translation parity; queue job IDs; published ports; the frontend host allow-list; model agreement between API and worker; stored photo addresses versus the deployment.
Tools	TypeScript through npm run build in every workspace; the i18n parity test; docker compose config; curl with a Host header; the smoke script.
Evidence this cycle	5 of 7 checks passed; CT-07 failed (finding F-01); CT-06 is not automated (risk R-07).
Left to other levels	Whether the behaviour behind a correct definition is itself correct — that is for unit and integration tests.

3.2.4 End-to-End Testing

End-to-end tests use the assembled, running system. Two kinds were executed: a recovery drill that kills a worker container during an import on the isolated stack, and a read-only smoke run against live data — once on the development stack and once on the EC2 deployment. The fifteen browser journeys need a person, real Firebase sign-in and real e-mail; they are written as numbered steps in Appendix C, and their results are entered in manual-results.csv as each one is performed. Automating them with Playwright is planned but was not built in this cycle.

Aspect	MotorX
Scope	E2E-A1 crash drill; E2E-A2 development-stack smoke; E2E-A3 EC2 smoke; E2E-01 to E2E-15 browser journeys covering sign-up, dealer onboarding, listings, CSV and ZIP import, discovery, comparison, stock tools, moderation, notifications, phones, languages and a Lighthouse audit.
Tools	Bash drill script; Python smoke script; Chrome, Edge and Firefox with DevTools device mode; Lighthouse; a real phone where available.
Evidence this cycle	3 system procedures executed (A1 passed; A2 16 of 17 checks; A3 15 of 17 checks); 0 of 15 browser journeys executed.

Aspect	MotorX
Left to other levels	Exhaustive input combinations (unit level) and sustained load (NF-03).

3.3 Testing Techniques and Types

Each type below uses the template's six fields, followed by its outcome in this cycle. The ISO/IEC 25010 characteristic it supports is given in brackets.

3.3.1 Data and Database Integrity Testing (*reliability; functional correctness*)

The API and the worker write to the same MongoDB collections, so neither process alone can be trusted to keep the data consistent. MotorX relies on the database for three guarantees: at most one open (draft or active) listing per normalised registration number, at most one listing per CSV row of an upload, and review decisions written in the same transaction as their audit entry. This type checks those guarantees where they are enforced.

Template field	MotorX application
Technique Objective	Confirm that vehicle, dealer, upload and audit records stay correct and unique however they are written — by a form, a CSV import, a bulk action, a retried job or two requests at the same moment.
Technique	Run repository and service code against the isolated replica set and read the collections back after each operation; insert the same plate in different spellings; replay the same batch and the same rejected row; take over an expired lease and try writing with the old owner; kill a worker mid-import (E2E-A1) and compare listings with distinct source rows; update photo URLs on keys that contain dots.
Oracles	The index definitions in listing.model.ts; expected counts computed from each fixture; the rule that sold and archived listings never block a relist; the drill's count of distinct source rows.
Required Tools	Vitest, Mongoose, the MongoDB 7 replica set in compose.test.yml, mongosh for inspection, the drill script.
Success Criteria	No duplicate open listing and no row imported twice in any test; counts after the drill equal the source rows; every audited action has its audit entry.
Special Considerations	The guard in apps/backend/src/test/db.ts refuses any database except motorx_test on a local host, so these tests cannot reach Atlas. Files run one at a time because each clears shared collections.
Outcome in this cycle	24 data-integration tests passed; the drill passed with 60,000 of 60,000 rows; one tooling defect found and fixed (F-08).

3.3.2 Function Testing (*functional suitability*)

The SRS sets out well over a hundred functional requirements for users, dealers, listings, uploads, ETL, search, notifications and administration. Many are enforced in more than one place — the same vehicle rules apply to the listing form, the CSV pipeline and bulk actions — so function testing checks each rule where it is implemented and then checks the whole use case through the API.

Template field	MotorX application
Technique Objective	Show that each implemented requirement gives the specified result for valid input and the specified refusal for invalid input, for every kind of user.
Technique	Equivalence classes and boundary values for vehicle attributes (six categories across the fuel types), prices, years, page sizes and upload files; state-transition tests for listing status and for applications (pending, rejected, pending again, approved); decision tables for bulk actions (which statuses each action may change); HTTP journeys that assert the stored result, not only the status code.
Oracles	The SRS v1.0 text; the shared Zod schemas; the allowed-transition table in listing.service.ts; docs/api-contract.md.
Required Tools	Vitest, supertest, Testing Library.
Success Criteria	Every executed case passes; every requirement area in Appendix G has an executed case or is shown as a gap.

Template field	MotorX application
Special Considerations	Behaviour beyond the SRS is tested to the same standard but reported as an extension, so it never inflates SRS coverage.
Outcome in this cycle	All automated function tests pass; confirmation in a browser is still outstanding (E2E-01 to E2E-11).

3.3.3 User Interface Testing (usability; compatibility)

One React application serves three audiences on very different screens: buyers on phones, dealers at desks with large inventories, and administrators. This cycle added a drawer menu for small screens, a bottom-sheet filter panel, swipeable photos, tables that become cards, and Sinhala and Tamil text, so the interface needs checking both for what it shows and for whether it can be operated.

Template field	MotorX application
Technique Objective	Confirm that every page can be reached, read and operated by touch, mouse and keyboard at phone, tablet and desktop widths, in all three languages, including its loading, empty and error states.
Technique	Component tests find elements the way assistive technology does — by role, label and name — and drive them with user-event: focus moving into and back out of the drawer and the price dialog, Escape to close, a swipe changing the photo while a vertical drag does not, translated labels. People then follow E2E-12 (widths 360, 390, 768 and 1440 px, keyboard only) and E2E-13 (native readers of Sinhala and Tamil).
Oracles	Visible labels and roles; WCAG 2.1 guidance on focus and target size (44 px) [5]; the translation dictionaries; screenshots of each state.
Required Tools	Vitest 5, jsdom, Testing Library, user-event; Chrome DevTools device mode; Lighthouse; a physical phone.
Success Criteria	All component tests pass; nothing scrolls sideways at 360 px; every control is reachable by keyboard; native readers accept the wording.
Special Considerations	jsdom does not apply CSS, so any claim about layout is accepted only from a real browser.
Outcome in this cycle	50 component tests passed; browser checks E2E-12, E2E-13 and E2E-15 are pending.

3.3.4 Performance Profiling (performance efficiency)

Two operations drive MotorX's cost: importing a large CSV (parsing, validating, embedding and inserting thousands of rows) and answering searches that combine filters, text and vectors. Profiling measures each one on its own, before any concurrency, so that later load results can be explained rather than only observed.

Template field	MotorX application
Technique Objective	Measure single-user response times of key API calls and the throughput of a 5,000-row import against PSR-01 to PSR-06.
Technique	A benchmark imports 5,000 generated car rows through the real worker pipeline into the replica set and MinIO, twice in a row; the smoke script times each request on the development stack and on the EC2 deployment.
Oracles	PSR-05 (about 5,000 records within 2 minutes); PSR-01 and PSR-02 (2 seconds); PSR-03 (5 seconds for semantic search).
Required Tools	Vitest benchmark (RUN_BENCHMARKS=1), smoke-script timings, docker stats.
Success Criteria	The benchmark finishes under 120 s with 5,000 of 5,000 rows; single API requests stay well under 2 s.
Special Considerations	Laptop measurements do not predict EC2 or Atlas behaviour; the embedding mode is recorded with each run.
Outcome in this cycle	5,000 rows in 11.1 s and 11.2 s (≈450 rows/s); single API requests on EC2 took 121–228 ms.

3.3.5 Load Testing (performance efficiency; reliability)

Buyers browse continuously, while dealers occasionally upload thousands of rows at once. The worker runs in its own

container, so an import should not slow browsing (PSR-06), but both share MongoDB Atlas, Redis and — on EC2 — one small machine. Load testing is where that separation is confirmed or disproved.

Template field	MotorX application
Technique Objective	Find how many simultaneous buyers the deployed stack serves within PSR-01 while an import runs, and at what number of users response times begin to climb steeply.
Technique	Planned k6 stages of 5, 20 and 50 virtual users, 10 minutes each after a warm-up, with a traffic mix of 70% browsing, 20% search and 10% vehicle details; a 5,000-row import is started during the 20-user stage; the rate limit is raised for the load generator's address for the duration.
Oracles	PSR-01 at the 95th percentile; error rate under 1% (proposed); memory, CPU and queue depth returning to normal after the run.
Required Tools	k6 [6], docker stats on the EC2 host, Atlas metrics.
Success Criteria	The agreed normal-load stage meets PSR-01, and the user count at which responses slow sharply is written down.
Special Considerations	Never run while others are demonstrating on the same stack. On burstable EC2 instance types, CPU credits must be recorded with each run or the results mislead.
Outcome in this cycle	Not executed (NF-03).

3.3.6 Security and Access Control Testing (security)

MotorX holds business documents (registration certificates, identity proof), commercially sensitive stock and personal contact details. Its main threats come from other users of the same system — a dealer probing a competitor's data, a suspended account still acting — and from hostile files uploaded as photos, CSVs or documents.

Template field	MotorX application
Technique Objective	Confirm that every protected action checks who is asking, which role they hold and whether they own the resource, and that uploaded content cannot carry hidden data or code to other users.
Technique	HTTP journeys act as a buyer, two dealers, an administrator and a suspended user and try each other's resources by ID; token tests cover caching, revocation and expiry; document tests use PDFs containing JavaScript, launch actions and embedded files, and programs renamed .pdf; image tests use GPS-tagged, rotated, oversized and fake images; rate-limit tests exceed budgets with Redis up and down; CI runs Gitleaks on the full history, Trivy on each image and npm audit; the EC2 security group and published ports are reviewed.
Oracles	PSR-08 to PSR-16; 401, 403, 404 or 429 with no data changed; OWASP ASVS 4.0.3 access-control and file-handling requirements as an external reference [7].
Required Tools	supertest, Vitest, sharp, Gitleaks, Trivy, npm audit; the AWS console for the security group.
Success Criteria	No read or write across dealers in any journey; unsafe files refused; no unaddressed high finding from the scanners; only SSH (from one address), 80, 443, 3000 and 4173 reachable on EC2.
Special Considerations	Firebase itself is replaced in the journeys; real tokens are exercised only by the smoke runs and the browser journeys. Multi-factor sign-in for administrators is not implemented and is recorded as an accepted risk (R-09).
Outcome in this cycle	64 security-related automated tests passed; protected routes returned 401 on EC2; the old deployment's exposure of Redis and MinIO was found and removed (F-14).

3.3.7 Failover and Recovery Testing (reliability)

MotorX's long jobs run in a worker that Docker may restart, a memory limit may kill, or a deployment may stop. Leases, checkpoints, a reaper and an e-mail outbox exist so that such interruptions neither lose nor repeat work — but those mechanisms only act when something fails, so this type makes things fail on purpose.

Template field	MotorX application
Technique Objective	Show that interrupted imports, lost queue messages, temporary storage or database errors and failed e-mails end either in a completed job with correct counts or in a clearly failed job the dealer can retry.

Template field	MotorX application
Technique	Kill a worker container mid-import (E2E-A1); simulate failures between batches, lease loss, exhausted attempts and lost queue messages with mocks (UT-20, UT-22, UT-24); make Redis unreachable while an upload is accepted (UT-07); fail SMTP five times in a row (UT-25). An unplanned outage also occurred on 26 Sep: Atlas became unreachable through DNS, the backend refused to start rather than run without its database, and it became healthy once name resolution returned.
Oracles	Recovery expectations in docs/RESILIENCE.md; row counts; job states; the retry schedule of 1 minute, 5 minutes, 30 minutes and 2 hours.
Required Tools	The drill script, Docker, Vitest fake timers.
Success Criteria	No row lost or repeated; every job ends in a terminal or resumable state.
Special Considerations	Drills run only on the isolated test stack. There is no second production instance to switch to, so this is restart-and-resume, not automatic failover.
Outcome in this cycle	The drill passed in 4 min 43 s; all recovery tests pass.

3.3.8 Configuration Testing (portability; compatibility)

MotorX now runs in four configurations: developers' laptops, an isolated test stack, GitHub Actions and a single EC2 instance behind an Elastic IP. A setting that is right in one can be wrong in another — ports bound to localhost, a development server that refuses unknown host names, photo addresses recorded with the host that was current at upload time. This type checks each configuration before and after deployment.

Template field	MotorX application
Technique Objective	Confirm the same code builds and behaves correctly in every configuration and browser, and that deployment-specific settings are right.
Technique	Build all workspaces on Node 24; run the suites in containers matching CI; resolve the Compose files and inspect port bindings; test the Vite host allow-list with and without its setting; run the smoke script against each deployed stack; open the site from its configured origin so CORS is exercised; record browser versions during the journeys.
Oracles	The Compose files, the .env keys each service reads, CI results and smoke results.
Required Tools	docker compose config, npm, GitHub Actions, the smoke script, curl.
Success Criteria	Builds pass; only ports 3000 and 4173 are public; the site loads by its EC2 host name; smoke runs pass on every stack.
Special Considerations	The EC2 stack and the developers' laptops share one Atlas database, so any data-changing check on either affects both (risk R-01).
Outcome in this cycle	Build, port and host checks passed; EC2 smoke 15 of 17 checks (F-01); CI exposed a test that only passed with real Firebase credentials (F-11, fixed).

3.4 Level × Technique Matrix

Each cell lists the cases that provide evidence for a test type at a level; “—” means that pairing is not used. Case details are in Appendices A to D.

Test type	Unit / component	Integration	Contract / consistency	End-to-end
3.3.1 Data integrity	UT-08, UT-20, UT-21	IT-01–IT-04, IT-07, IT-11	CT-06	E2E-A1, E2E-06
3.3.2 Function	UT-03–UT-08, UT-12–UT-14, UT-16–UT-19	IT-08–IT-10	CT-01	E2E-01–E2E-11, E2E-14
3.3.3 User interface	UT-28–UT-40	—	CT-02, CT-05	E2E-12, E2E-13, E2E-15
3.3.4 Performance profiling	—	NF-01	—	E2E-A2, E2E-A3, NF-02
3.3.5 Load	—	—	—	NF-03

Test type	Unit / component	Integration	Contract / consistency	End-to-end
3.3.6 Security	UT-01, UT-02, UT-05, UT-09, UT-10, UT-15, UT-28, UT-30	IT-05, IT-06, IT-08	CT-04	E2E-02, E2E-11, E2E-A3, NF-05
3.3.7 Failover and recovery	UT-07, UT-20, UT-22–UT-26	IT-07	CT-03	E2E-A1
3.3.8 Configuration	UT-02	—	CT-01, CT-04, CT-05, CT-07	E2E-A2, E2E-A3, NF-05

Three observations follow from the matrix. Most of the executed evidence sits in the unit and integration columns, which is where MotorX’s riskiest rules — plate uniqueness, lease ownership, dealer isolation — can be tested precisely. Load is the only type with nothing executed at any level, which makes the number of simultaneous users the system can serve the largest open question in this report. And the thinnest column, contract and consistency, found or confirmed three of the four deployment problems of this cycle (F-01, F-12, F-14), which argues for extending it — CT-06 in particular — before the ECS path is used.

4. Test Case Inventory

The cases below are grouped by the part of MotorX they exercise; a case can appear under more than one heading. The result in brackets is the latest recorded; full specifications are in the appendices.

4.1 Identity and access

- UT-01 — Test database target safety (passed)
- UT-02 — Production database URI guard (passed)
- UT-15 — Token verification cache (passed)
- UT-28 — Role-protected pages (passed)
- UT-29 — Email verification banner (passed)
- UT-30 — Per-account data isolation (passed)
- IT-06 — Rate limits with real Redis (passed)
- IT-08 — Access control across dealers and roles (passed)
- IT-12 — Real Firebase identity (not run)
- E2E-01 — Buyer registration, sign-in and session (not run)
- E2E-11 — Administrator moderation and suspension (not run)

4.2 Dealer onboarding and profiles

- UT-04 — Dealer application validation (passed)
- UT-05 — Dealer document content checks (passed)
- UT-26 — Document retention (passed)
- UT-31 — Dealer application from an existing account (passed)
- UT-32 — Dealer profile editing (passed)
- IT-01 — Dealer application persistence (passed)
- IT-02 — Pending applications oldest first (passed)
- IT-05 — Audited document access (passed)
- IT-09 — Dealer lifecycle and admin monitoring (passed)
- E2E-02 — Dealer application and approval (not run)
- E2E-03 — Rejection, correction and resubmission (not run)

4.3 Listings and photos

- UT-08 — Listing validation and identity (passed)
- UT-09 — Image signature check (passed)
- UT-10 — Image re-encoding (passed)
- UT-11 — Photo migration and small copies (passed)
- IT-03 — Registration uniqueness in MongoDB (passed)
- IT-04 — Listing update persistence (passed)
- IT-11 — Photo URL updates in MongoDB (passed)
- CT-07 — Stored photo addresses match the deployment (failed)
- E2E-04 — Create, edit, publish and photograph a vehicle (not run)

- E2E-05 — Recover from a failed photo upload (not run)

4.4 Buyer discovery

- UT-12 — Search query analysis (passed)
- UT-13 — Vehicle similarity scoring (passed)
- UT-14 — Pagination metadata (passed)
- UT-36 — Vehicle page on phones (passed)
- UT-37 — Compare vehicles (passed)
- IT-10 — Bulk tools, stale stock and discovery (passed)
- IT-15 — Search relevance on a labelled dataset (not run)
- E2E-07 — Buyer search, filters and dealer contact on a phone (not run)
- E2E-08 — Similar vehicles, recommendations and compare (not run)
- NF-06 — Search relevance on labelled queries (not run)

4.5 Inventory import and the worker

- UT-06 — CSV upload validation (passed)
- UT-07 — Upload acceptance and controlled retry (passed)
- UT-16 — CSV extraction (passed)
- UT-17 — Vehicle normalization (passed)
- UT-18 — Batch transformation (passed)
- UT-19 — Category and powertrain rules (passed)
- UT-20 — CSV ETL orchestration (passed)
- UT-21 — ZIP photo processing (passed)
- UT-22 — Job lease renewal (passed)
- UT-23 — Retry classification (passed)
- UT-24 — Lost-job reconciliation (passed)
- UT-41 — Queue publishing for repeat photo uploads (passed)
- IT-07 — Lease ownership and idempotent writes (passed)
- IT-14 — CSV and ZIP through queue and live worker (not run)
- CT-03 — Queue job IDs agree between API and worker (passed)
- E2E-A1 — Worker killed during a 60,000-row import (passed)
- E2E-06 — CSV import, corrections and ZIP photos (not run)
- NF-01 — 5,000-row import throughput (passed)

4.6 Dealer stock tools

- UT-27 — Stale-stock reminders (passed)
- UT-33 — Upload details and publish-all (passed)
- UT-34 — Bulk and stale-stock tools (passed)
- IT-10 — Bulk tools, stale stock and discovery (passed)
- E2E-09 — Bulk actions and stale stock (not run)

- E2E-10 — Stale-stock reminder notification (not run)

4.7 Administration and notifications

- UT-03 — Admin request validation (passed)
- UT-25 — Email outbox delivery (passed)
- UT-35 — Admin approvals, dashboard and monitoring (passed)
- IT-02 — Pending applications oldest first (passed)
- IT-05 — Audited document access (passed)
- IT-09 — Dealer lifecycle and admin monitoring (passed)
- IT-13 — Real SMTP delivery and failure (not run)
- E2E-11 — Administrator moderation and suspension (not run)
- E2E-14 — Notification centre and email status (not run)

4.8 Phones, languages and accessibility

- UT-36 — Vehicle page on phones (passed)
- UT-38 — Mobile navigation and card tables (passed)
- UT-39 — Sinhala and Tamil (passed)
- UT-40 — WhatsApp number formatting (passed)
- CT-02 — Sinhala and Tamil dictionaries agree with English (passed)
- E2E-07 — Buyer search, filters and dealer contact on a phone (not run)
- E2E-12 — Responsive layout and keyboard use (not run)
- E2E-13 — Sinhala and Tamil (not run)
- E2E-15 — Mobile performance audit (not run)

4.9 Shared contracts, configuration and deployment

- CT-01 — Shared contracts compile into every consumer (passed)
- CT-04 — Deployment publishes only the website and the API (passed)
- CT-05 — Frontend accepts the address users open (passed)
- CT-06 — API and worker agree on shared collection shapes (not run)
- CT-07 — Stored photo addresses match the deployment (failed)
- E2E-A2 — Read-only smoke of the development stack (failed)
- E2E-A3 — Read-only smoke of the EC2 deployment (failed)
- NF-02 — Response-time percentiles (not run)
- NF-03 — Concurrent buyers during an import (not run)
- NF-04 — Upload limits and unsafe files (passed)
- NF-05 — Build and supply-chain checks (passed)

5. Deliverables

The artefacts below are produced by this test effort and are what its success is judged on. Paths are relative to the repository root.

- This report: docs/MotorX_Test_Plan_Report.docx (Word, on the supplied template) and docs/MotorX_Test_Plan_Report.md.
- Case register: docs/MotorX_Test_Case_Register.csv — all 94 cases with level, requirement, steps, expected and actual result, tester, date and finding.
- Executed-test register: docs/test-evidence/automated-test-register.csv — every automated test (348) with its file, status and duration.
- Raw results: docs/test-evidence/backend-results.json, worker-results.json and frontend-results.json (Vitest JSON).
- Performance and smoke evidence: benchmark-output.txt, live-smoke-output.txt, live-smoke-results.json and ec2-smoke-results.json.
- Configuration evidence: compose-ec2-ports.txt and vite-allowed-hosts.txt.
- Code coverage: coverage-backend.json, coverage-worker.json and coverage-frontend.json (v8 summaries per file).
- Build and environment: build-output.txt and run-metadata.json.
- Manual results sheet: docs/test-evidence/manual-results.csv, filled in during browser journeys, with screenshots under test-evidence/screenshots and Lighthouse reports under test-evidence/lighthouse.
- Test tooling: scripts/drills/kill-worker-mid-import.sh, scripts/smoke/live-stack-smoke.py and the report generator, which is kept outside the repository.
- Defect log and traceability matrix: Appendices F and G.

5.1 Test Evaluation Summaries

The summary is regenerated after every test session from the registers, so the figures always match the evidence. “Not run” items never count towards the pass rate, which is $\text{passed} \div \text{executed}$.

Category	Total	Executed	Passed	Failed / blocked	Not run	Pass rate
Unit and component tests	291	291	291	0	0	100%
Integration tests — data (IT-01–IT-07, IT-11)	24	24	24	0	0	100%
Integration tests — HTTP journeys (IT-08–IT-10)	32	32	32	0	0	100%
Integration cases — external services (IT-12–IT-15)	4	0	0	0	4	—
Contract and consistency checks (CT-01–CT-07)	7	6	5	1	1	83%
End-to-end — system procedures (E2E-A1–A3)	3	3	1	2	0	33%
End-to-end — browser journeys (E2E-01–E2E-15)	15	0	0	0	15	—
Performance (NF-01, NF-02)	2	1	1	0	1	100%
Load (NF-03)	1	0	0	0	1	—
Security and supply chain (NF-04, NF-05)	2	2	2	0	0	100%
Search quality (NF-06)	1	0	0	0	1	—
Total	382	359	356	3	23	99%
Security view — subset of the rows above, not added to the total	64	64	64	0	0	100%

Rows count automated tests where tests exist and cases or checks elsewhere, so the total mixes the two and is best read row by row. Every failure recorded so far has the same cause: photos whose files live only on another machine (F-01).

Assessment: the automated evidence is broad and fully green, the EC2 deployment is healthy with its internal services closed to the internet, and every tested recovery path works. The build is not yet ready for full acceptance for three reasons: 15 of the 15 browser journeys have not been executed, F-01 is only partly fixed, and the EC2 site is still served over plain HTTP (F-15). Load (NF-03) and response-time percentiles (NF-02) are the largest measurement gaps; if they cannot be run in time they will be reported as unmet rather than estimated.

5.1.1 Test Logs

The outputs of the testing tools are recorded here as each tool is run, starting with the test logs. Each workspace's Vitest suite was run in a terminal: the backend and worker inside the isolated test stack (MongoDB replica set, Redis and MinIO in Docker, never Atlas) and the frontend on the developer laptop. Every run ended green, and the totals match the executed-test register: $147 + 94 + 107 = 348$ tests.

Workspace	Command	Test files	Tests	Result
Backend	<code>docker compose -f compose.yml -f compose.test.yml run --rm backend</code>	26 passed	147 passed	Passed
Worker	<code>docker compose -f compose.yml -f compose.test.yml run --rm worker</code>	13 passed, 1 skipped	93 passed, 1 skipped	Passed
Frontend	<code>npm run test --workspace @motorx/frontend</code>	20 passed	107 passed	Passed

```

at processTicksAndRejections (node:internal/process/task_queues:104:5)
at file:///app/node_modules/express-rate-limit/dist/index.mjs:836:5

stderr | src/shared/middleware/rateLimits.test.ts > rate limits > lets requests through when the shared store (Redis) is down
express-rate-limit: error from store, allowing request without rate-limiting. Error: connect ECONNREFUSED
at Object.increment (/app/apps/backend/src/shared/middleware/rateLimits.test.ts:34:38)
at file:///app/node_modules/express-rate-limit/dist/index.mjs:889:52
at processTicksAndRejections (node:internal/process/task_queues:104:5)
at file:///app/node_modules/express-rate-limit/dist/index.mjs:836:5

✓ src/shared/middleware/rateLimits.test.ts (5 tests) 168ms
✓ src/modules/marketplace/listing.repository.test.ts (6 tests) 143ms
✓ src/modules/admin/admin.documentAccess.test.ts (3 tests) 146ms
✓ src/shared/utls/imageReencode.test.ts (7 tests) 779ms
  ✓ refuses an image over the 40 megapixel cap even though the file is small 432ms
  ✓ keeps PNG documents as PNG and shrinks them to the requested size 303ms
✓ src/modules/marketplace/listing.service.test.ts (2 tests) 239ms
✓ src/modules/dealers/dealerDocument.content.test.ts (9 tests) 31ms
✓ src/shared/middleware/verifyFirebaseToken.test.ts (4 tests) 12ms
✓ src/scripts/listingImageMigration.db.test.ts (2 tests) 144ms
✓ src/test/db.safety.test.ts (8 tests) 14ms
✓ src/modules/buyers/buyer.similarity.test.ts (5 tests) 7ms
✓ src/modules/inventory/inventory.validation.test.ts (6 tests) 10ms
✓ src/modules/search/search.queryAnalyzer.test.ts (5 tests) 23ms
✓ src/modules/inventory/inventory.queue.test.ts (4 tests) 10ms
✓ src/config/mongoUri.test.ts (10 tests) 12ms
✓ src/modules/dealers/dealer.validation.test.ts (3 tests) 13ms
✓ src/modules/dealers/dealer.repository.test.ts (1 test) 89ms
✓ src/modules/admin/admin.repository.test.ts (1 test) 144ms
✓ src/modules/admin/admin.validation.test.ts (4 tests) 10ms
✓ src/modules/marketplace/listingImage.service.test.ts (2 tests) 4ms
✓ src/shared/utls/pagination.test.ts (2 tests) 4ms

Test Files 26 passed (26)
Tests 147 passed (147)
Start at 19:59:01
Duration 16.39s (transform 936ms, setup 0ms, import 7.19s, tests 4.86s, environment 5ms)

PS D:\MotorX\motorx> |

```

Figure 1: Backend test log (Vitest 4.1.10): 26 test files, 147 tests passed

```

RUN v4.1.10 /app/apps/worker

✓ src/services/imageProcessing.service.test.ts (17 tests) 1487ms
  ✓ skips an image above the pixel cap even when the file itself is small 409ms
  ✓ shrinks large photos to the maximum stored dimension 467ms
  ✓ also stores an 800 px small copy for cards and phones, and links it on the listing 452ms
✓ src/services/uploadJob.service.test.ts (9 tests) 61ms
✓ src/pipeline/validate.test.ts (17 tests) 22ms
(node:84) [MONGOOSE] Warning: 'errors' is a reserved schema pathname and may break some functionality.
To disable this warning pass 'suppressReservedKeysWarning' as a schema option.
(Use 'node --trace-warnings ...' to show where the warning was created)
✓ src/repositories/jobSafety.db.test.ts (4 tests) 161ms
↓ src/benchmarks/importThroughput.bench.test.ts (1 test | 1 skipped)
✓ src/jobs/emailOutbox.job.test.ts (7 tests) 18ms
✓ src/pipeline/normalize.test.ts (6 tests) 8ms
✓ src/jobs/reaper.job.test.ts (6 tests) 17ms
✓ src/jobs/documentRetention.job.test.ts (3 tests) 18ms
✓ src/jobs/staleListingReminder.job.test.ts (3 tests) 9ms
✓ src/pipeline/transform.test.ts (3 tests) 23ms
✓ src/services/jobLease.test.ts (4 tests) 13ms
✓ src/services/transientError.test.ts (12 tests) 6ms
✓ src/pipeline/extract.test.ts (2 tests) 16ms

Test Files 13 passed | 1 skipped (14)
Tests 93 passed | 1 skipped (94)
Start at 18:32:20
Duration 7.62s (transform 531ms, setup 0ms, import 3.26s, tests 1.86s, environment 3ms)

PS D:\MotorX\motorx>

```

Figure 2: Worker test log (Vitest 4.1.10): 93 tests passed, the 5,000-row benchmark skipped

```

RUN v5.0.2 D:/MotorX/MotorX/apps/frontend

✓ src/portals/buyer/pages/VehicleDetails.test.tsx (4 tests) 500ms
✓ src/portals/dealer/pages/ListingManager.test.tsx (5 tests) 780ms
✓ src/features/auth/pages/RegisterPage.test.tsx (3 tests) 465ms
✓ src/shared/components/mobileLayout.test.tsx (3 tests) 496ms
✓ src/features/compare/compare.test.tsx (4 tests) 586ms
  ✓ compare vehicles (4)
    ✓ collects up to three vehicles, says when the list is full, and opens the comparison 447ms
✓ src/portals/dealer/pages/UploadDetails.test.tsx (5 tests) 531ms
✓ src/shared/i18n/i18n.test.tsx (4 tests) 386ms
  ✓ Sinhala and Tamil support (4)
    ✓ switches the whole site language, sets the page lang, and remembers the choice 355ms
✓ src/portals/dealer/pages/DealerProfile.test.tsx (3 tests) 543ms
  ✓ DealerProfile (3)
    ✓ saves edited details and confirms 367ms
stderr | src/features/auth/components/EmailVerificationBanner.test.tsx > EmailVerificationBanner >
Warning: The current testing environment is not configured to support act(...)

stderr | src/features/auth/components/EmailVerificationBanner.test.tsx > EmailVerificationBanner >
Warning: The current testing environment is not configured to support act(...)
Warning: The current testing environment is not configured to support act(...)

✓ src/features/auth/components/EmailVerificationBanner.test.tsx (4 tests) 385ms
✓ src/portals/admin/pages/adminPages.test.tsx (6 tests) 404ms
✓ src/features/auth/components/RoleGuard.test.tsx (4 tests) 58ms
✓ src/features/auth/context/AuthProvider.test.tsx (2 tests) 177ms
✓ src/shared/utis/phone.test.ts (3 tests) 5ms

Test Files 13 passed (13)
Tests 50 passed (50)
Start at 00:01:37
Duration 12.72s (environment 54%, tests 24%, setup 12%, import 6%, transform 4%)

```

Figure 3: Frontend test log (Vitest 5.0.2) before the new component tests were added: 13 test files, 50 tests passed

The frontend screenshot was taken before seven component test files were added on 28 September 2026 for the listing form, CSV upload and dealer dashboard, admin pages, sign-in, notifications, marketplace and the API services (UT-42 to UT-48). The suite now has 107 tests in 20 files, all passing, as recorded in frontend-results.json.

The one skipped worker test is the 5,000-row import benchmark (NF-01), which runs only when RUN_BENCHMARKS=1 because it takes several seconds against a real database; its separate runs are recorded in benchmark-output.txt. The messages printed in between are expected and do not fail any test: the rate-limit test switches Redis off on purpose to prove requests still get through (ECONNREFUSED), Mongoose warns that one schema has a field named “errors”, and React’s test renderer prints act() warnings for the e-mail verification banner.

5.1.2 User Interface Test Details

User-interface journeys were recorded and replayed with the Recorder panel of Chrome DevTools, which records what a user clicks and types and then plays it back against the live site, in the way the Selenium IDE is used for web testing. The journey “motorx vehicle comparison” was recorded on the EC2 deployment and replayed on the desktop profile (973 × 908 px, no throttling, 5-second timeout per step).

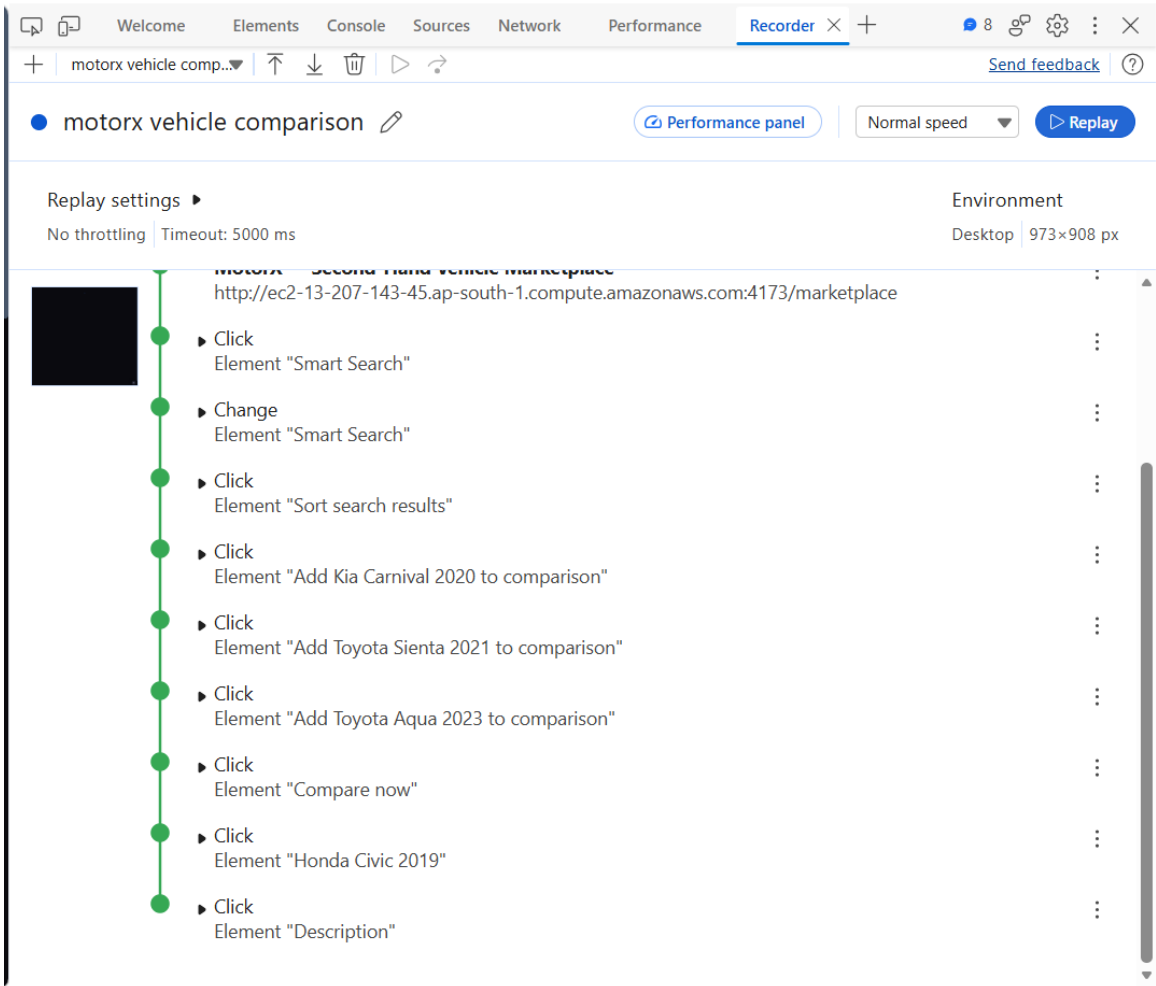


Figure 4: Chrome DevTools Recorder: the “motorx vehicle comparison” journey replayed on the EC2 site, every step completed

Step	Action	What it checks
1	Open /marketplace	The marketplace loads on the public site
2–3	Click and type in Smart Search	The natural-language search box accepts a query
4	Change the sort order	Sorting control works on the results
5–7	Add Kia Carnival 2020, Toyota Sienta 2021 and Toyota Aqua 2023 to comparison	Compare toggles on the result cards, found by their accessible labels
8	Click Compare now	The compare tray opens the comparison page
9–10	Open Honda Civic 2019 and its Description	Navigation to a vehicle and its details

Every step completed on replay (green), so the buttons, fields and links of this journey can be found and used by their visible names and labels, which also indicates that they are exposed correctly to assistive technology. The recording can be exported from the Recorder and replayed after each deployment as a quick regression check. It supports the buyer journeys E2E-07 and E2E-08 but does not replace them: those also cover a phone layout, the contact bar and recommendations, and their results stay as recorded in Appendix C.

5.1.3 Code Inspection and Dependency Audit

For code inspection, the TypeScript compiler checks all three workspaces for type errors, and npm audit checks the

production dependencies against the public advisory database. Both were run on 28 September 2026 from the repository root. The screenshot shows the start of the output; the full output, including the third advisory (uuid), is kept in test-evidence/typecheck-and-audit.txt.

```

1 PS D:\MotorX\motorx> npx tsc -p apps/backend/tsconfig.json --noEmit
2
3 PS D:\MotorX\motorx> npx tsc -p apps/worker/tsconfig.json --noEmit
4
5 PS D:\MotorX\motorx> npx tsc -p apps/frontend/tsconfig.json --noEmit
6
7 PS D:\MotorX\motorx> npm audit --omit=dev
8 # npm audit report
9
10 csv-parse <7.0.2
11 Severity: moderate
12 node-csv: Prototype replacement still reachable via columns path - https://github.com/advisories/GHSA-8cw4-87c7-c6xx
13 fix available via 'npm audit fix --force'
14 Will install csv-parse@7.0.3, which is a breaking change
15 node_modules/csv-parse
16
17 qs 2.2.5 - 6.15.3
18 Severity: moderate
19 qs array-limit bypass via bracket-key comma parsing - https://github.com/advisories/GHSA-x5fp-wj9c-mxmx
20 qs: Denial of Service via Attacker Controlled isBuffer - https://github.com/advisories/GHSA-4mjr-xmp4-gh2g
21 fix available via 'npm audit fix'
22 node_modules/qs
23 body-parser 1.20.5 - 1.20.6
24 Depends on vulnerable versions of qs
25 node_modules/body-parser
26 express 4.22.2
27 Depends on vulnerable versions of qs
28 node_modules/express

```

Figure 5: Type-check of the three workspaces (no output means no errors) and the start of npm audit for the production dependencies

Result: the three type-checks printed nothing, so the backend, worker and frontend compile without type errors. The audit found 12 moderate advisories and no high or critical ones, which meets the NF-05 criterion of no unaddressed high or critical vulnerability. The 12 come from three packages; the other nine entries are packages that depend on them. The table records whether MotorX reaches each one.

Package	Advisory	Does MotorX reach it?	Action
csv-parse 5.6.0 (worker)	Prototype replacement through the columns option (GHSA-8cw4-87c7-c6xx)	Yes. The worker reads dealer CSV files with columns: true, so the header row of an uploaded file reaches the affected code. Only approved, signed-in dealers can upload, which limits who could try.	Upgrade to csv-parse 7.x (a major version) or reject reserved header names such as __proto__, with a worker test (F-18).
qs 6.15.3, through Express 4.22.2 and body-parser (backend)	Array-limit bypass and a denial of service (GHSA-x5fp-wj9c-mxmx, GHSA-4mjr-xmp4-gh2g)	Yes. Express parses the query string of every request with qs, including the public search endpoints.	Run npm audit fix, which updates qs within 6.x, then rerun the backend tests (F-18).
uuid 9.0.1, inside firebase-admin 13.10.0	Missing buffer bounds check when a caller passes its own buffer (GHSA-w5hq-g745-h8pq)	Not directly. MotorX code does not import uuid; it is used inside the Google Cloud client libraries that firebase-admin depends on.	Upgrade firebase-admin to 14.5 (a major version) in a later cycle.

5.1.4 Browser Performance (Lighthouse and Network)

Lighthouse in Chrome DevTools was run twice against the EC2 deployment. The first run audited the marketplace page while EC2 served the frontend from the Vite development server. The second, on 28 September 2026, audited the home page after EC2 was switched to the production build, served by nginx with compression and long-term caching of the built files (compose.ec2.yml, infrastructure/nginx/nginx.conf).

Run	Page	Frontend served by	Performance	Accessibility	Best practices	SEO
Before	/marketplace	Vite development server	26	94	56	92

Run	Page	Frontend served by	Performance	Accessibility	Best practices	SEO
After	/ (home)	Production build on nginx	84	94	56	92

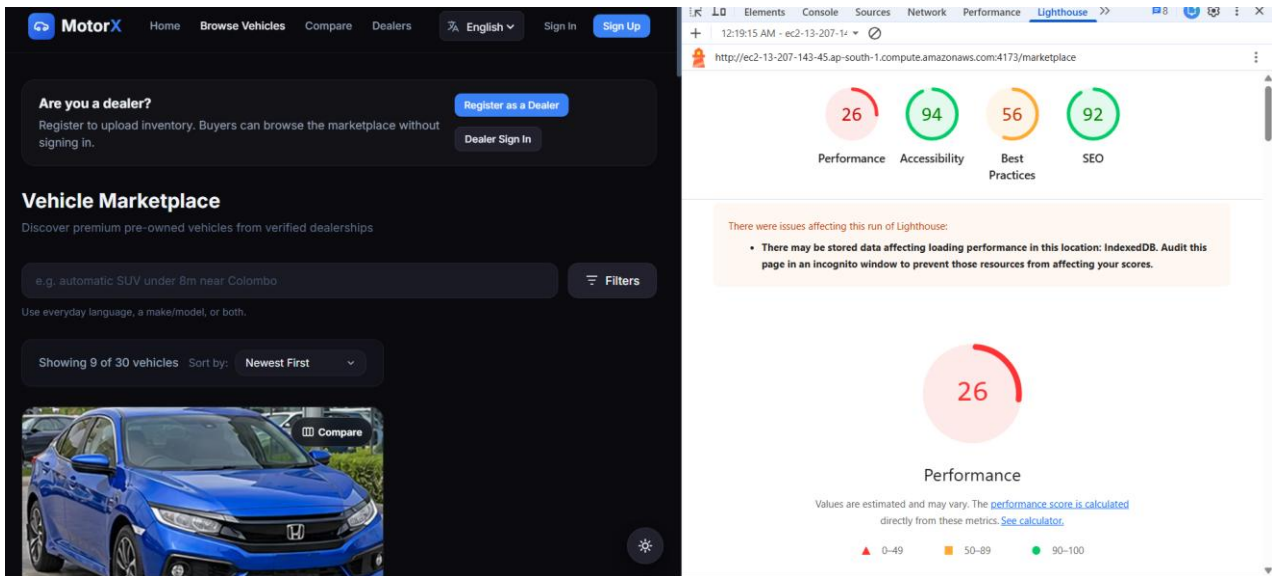


Figure 6: Lighthouse before: marketplace page served by the Vite development server (Performance 26)

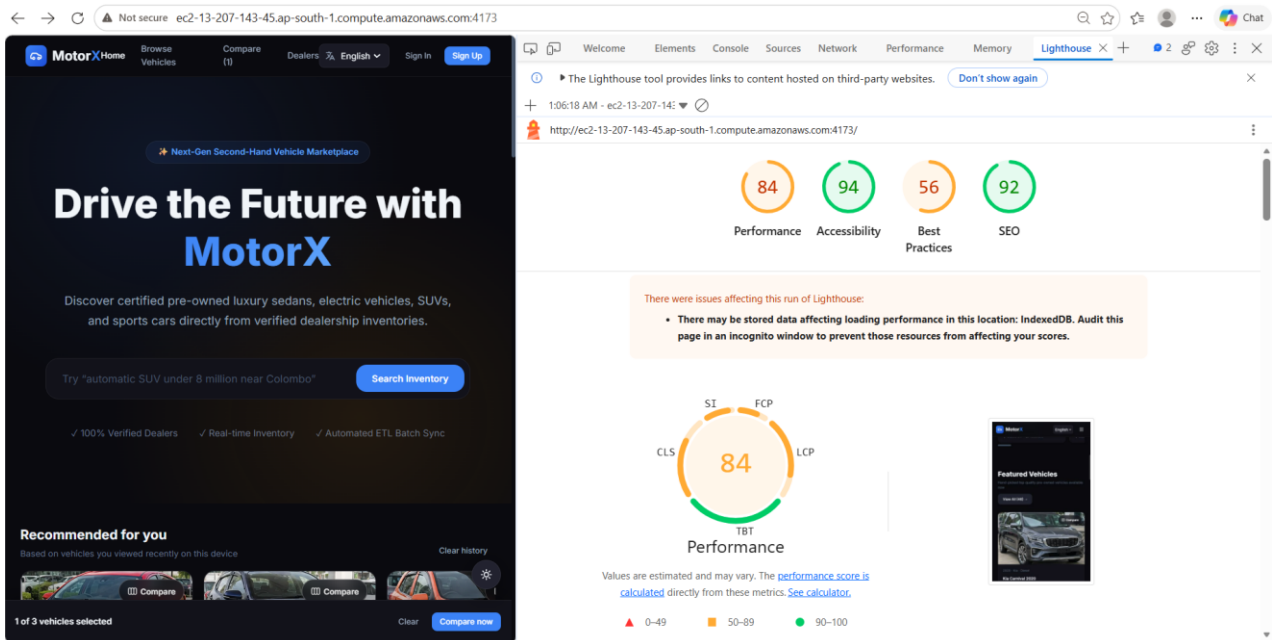


Figure 7: Lighthouse after: home page served from the production build on nginx (Performance 84)

Performance rose from 26 to 84. The main cause is the change of server: the development server sends hundreds of separate, uncompressed modules, while the production build sends a few compressed bundles that the browser can cache. The two runs audited different pages, so the comparison is indicative; auditing the marketplace and a vehicle page on the production build completes E2E-15, which stays open until then. Both runs warned that data stored in the browser (IndexedDB, used by Firebase sign-in) may have lowered the scores; an incognito window avoids this. Best practices stays at 56, largely because the site is still served over plain HTTP (F-15). Accessibility and SEO did not change.

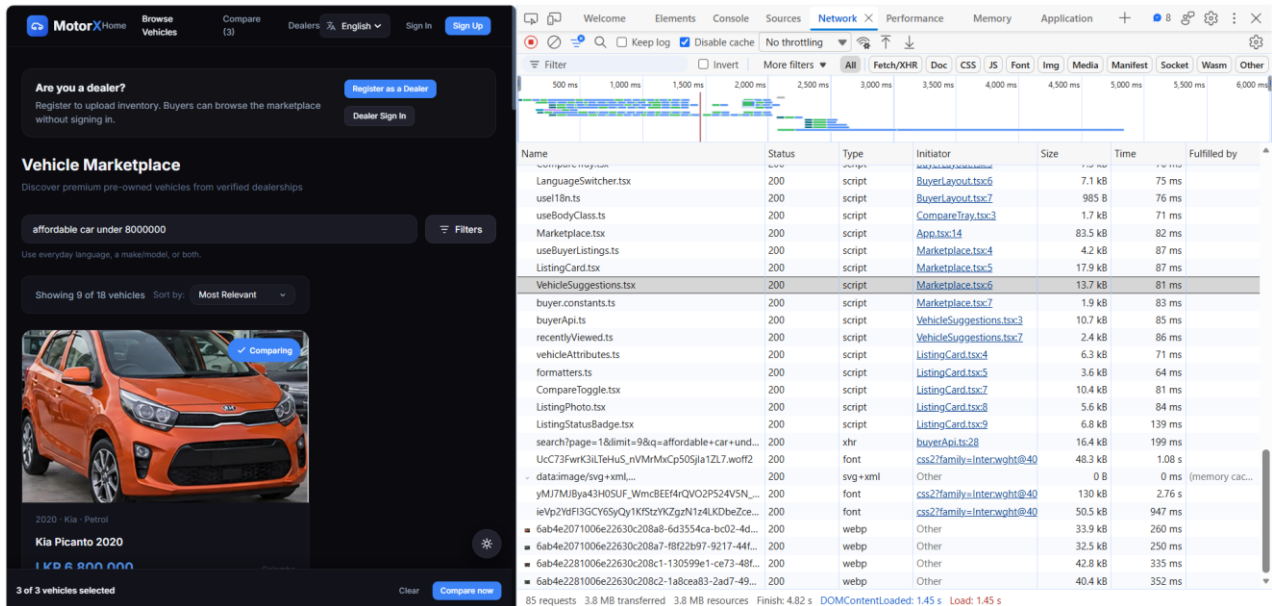


Figure 8: Chrome DevTools Network panel for a marketplace search with the cache disabled, captured on the development server

Figure 8 was captured with the cache disabled while searching for “affordable car under 8000000”: 85 requests, 3.8 MB transferred, DOMContentLoaded and load at 1.45 s, and the last request finished at 4.82 s. The request list names individual source files (.tsx), so it was taken on the development server, before the production build. The search API call returned in 199 ms and each listing photo, served as WebP, in 250–352 ms. The slowest items are the three Inter font files from Google Fonts (48–130 kB, up to 2.76 s); hosting the font with the site, or loading only the weights in use, would shorten the first load.

5.1.5 Server Monitoring

The running EC2 deployment was inspected over SSH on 28 September 2026 with three commands: docker compose ps (which services run and whether their health checks pass), docker stats (processor and memory use per container) and a request to the backend readiness endpoint.

```

ubuntu@ip-172-31-3-37: ~/m
Container motorx-minio-init-1 Exited 0.7s
ubuntu@ip-172-31-3-37:~/motorx$ docker compose -f compose.yml -f compose.ec2.yml ps
NAME                SERVICE          IMAGE                               STATUS          PORTS          COMMAND
motorx-backend-1    backend          motorx-backend                     Up 6 hours (healthy)  0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp  "/docker-entrypoint
motorx-frontend-1   frontend         sha256:5618489439cfb594f4ebelebb14493929890f45e45ce5817900300de19b2ea77  Up 35 minutes (healthy)  0.0.0.0:4173->80/tcp, [::]:4173->80/tcp      "/docker-entrypoi
motorx-minio-1      minio            quay.io/minio/minio:latest        Up 6 hours          127.0.0.1:9000-9001->9000-9001/tcp          "/usr/bin/docker-
motorx-redis-1      redis            redis:7-alpine                     Up 6 hours (healthy)  127.0.0.1:6379->6379/tcp                  "docker-entrypoint
motorx-worker-1     worker          motorx-worker                       Up 6 hours (healthy)
ubuntu@ip-172-31-3-37:~/motorx$ docker stats --no-stream
CONTAINER ID   NAME                CPU %     MEM USAGE / LIMIT   MEM %     NET I/O       BLOCK I/O
919de69e2f10   motorx-frontend-1   0.00%     9.141MiB / 1.861GiB  0.48%     69.1kB / 622kB  6.34MB / 8.19kB
5467543d10fa   motorx-backend-1    0.10%     79.28MiB / 1.861GiB  4.16%     77.1MB / 39MB   51.3MB / 20.4MB
d08be9ff4d8a   motorx-worker-1     0.09%     101.9MiB / 1.861GiB  5.35%     23.7MB / 13.2MB 31.9MB / 20.6MB
665ccf2eb705   motorx-redis-1      0.46%     6.461MiB / 1.861GiB  0.34%     4.63MB / 12.6MB 15.3MB / 5.77MB
ce159995839e   motorx-minio-1      0.06%     122MiB / 1.861GiB    6.41%     20.2MB / 45.3MB 153MB / 111MB
ubuntu@ip-172-31-3-37:~/motorx$ ^[[200~curl -s http://localhost:3000/health/ready; echo~
curl: command not found
Command 'echo~' not found, did you mean:
  command 'echo' from deb coreutils-from-gnu (0.0.0~ubuntu25)
  command 'echo' from deb coreutils-from-uutils (0.0.0~ubuntu25)
  command 'echo' from deb coreutils-from-busybox (0.0.0~ubuntu25)
  command 'echo' from deb coreutils-from-toybox (0.0.0~ubuntu25)
Try: sudo apt install <deb name>
ubuntu@ip-172-31-3-37:~/motorx$ curl -s http://localhost:3000/health/ready; echo
{"success":true,"data":{"service":"backend","status":"READY","dependencies":{"database":"ready","redis":"ready"}},
{"meta":null}}

```

Figure 9: EC2 server monitoring: container status, resource use per container and the backend readiness check

Container	Status	Published on	CPU	Memory
frontend (nginx, production build)	Up, healthy	0.0.0.0:4173 → 80	0.00%	9.1 MiB
backend (API)	Up, healthy	0.0.0.0:3000	0.10%	79.3 MiB
worker	Up, healthy	not published	0.09%	101.9 MiB
redis	Up, healthy	127.0.0.1:6379 only	0.46%	6.5 MiB
minio	Up	127.0.0.1:9000–9001 only	0.06%	122 MiB

All five services were running and every health check passed; the readiness endpoint answered READY with the database and Redis both ready. Together the containers used about 320 MiB of the instance's 1.86 GiB and almost no processor time while idle, so the instance has room for normal use; the heavy moment is building images, which is why they are built one at a time (F-13). The output also confirms two earlier results on the live server: the website is now served by nginx from the production build (port 4173 maps to 80, R-05), and Redis and MinIO accept connections only from the machine itself (F-14). The first attempt at the readiness request in the screenshot failed only because the command was pasted with stray characters; the retry below it is the result.

5.1.6 Database Performance

The shared MongoDB Atlas cluster was examined in two ways on 27–28 September 2026: MongoDB Compass explained the query behind the marketplace's default view, and the Atlas monitoring page showed the load on the cluster over one hour.

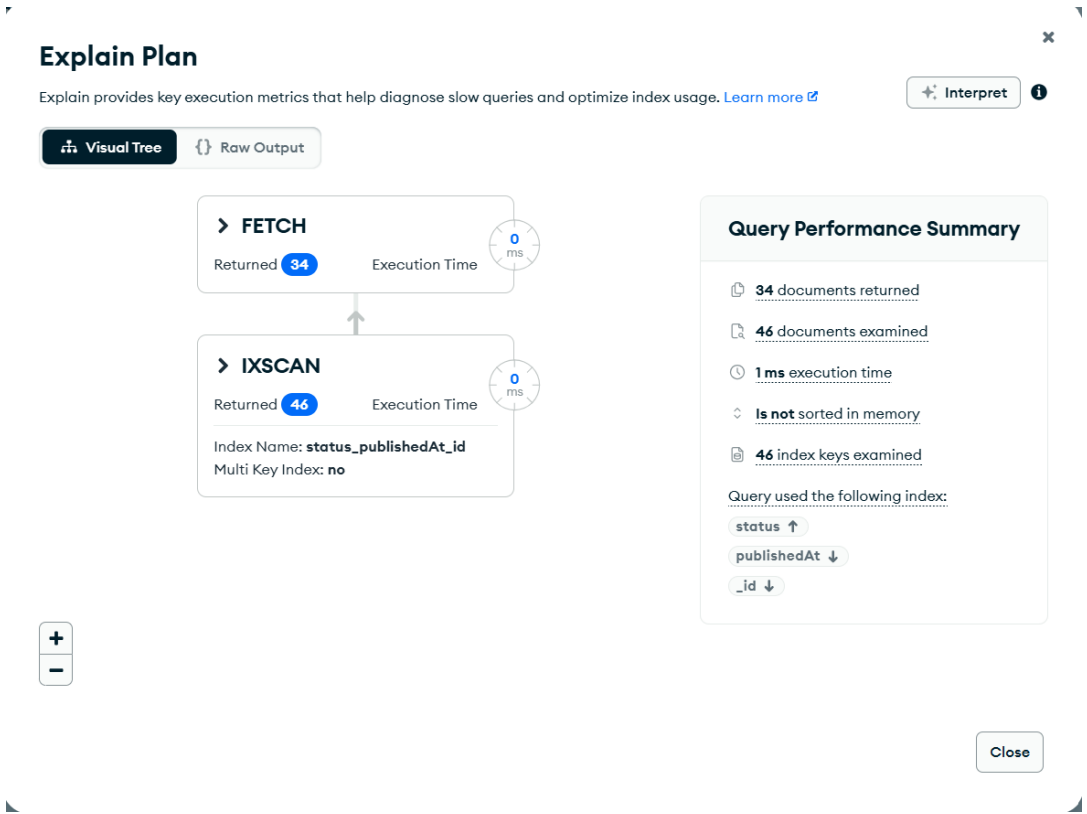


Figure 10: Compass explain plan for active cars, newest first: an index scan on status_publishedAt_id, 34 documents returned in 1 ms

The query was the one the marketplace sends by default, with filter { status: "active", category: "car" } and sort { publishedAt: -1 }. MongoDB answered it from the status_publishedAt_id index (IXSCAN) rather than reading the whole collection: it examined 46 index keys and 46 documents of the 113 in the collection, returned 34, and took 1 ms. Because the index already holds listings in publishing order, no sort was done in memory. The 12 extra documents examined are active listings of other vehicle types, which are filtered out after they are read; an index on status, category and publishedAt together would remove that step, but at this data size it is not worth the extra write cost.

Measure	Result
Plan	IXSCAN on status_publishedAt_id, then FETCH
Index keys / documents examined	46 / 46 (collection holds 113)
Documents returned	34
Execution time	1 ms
Sorted in memory	No

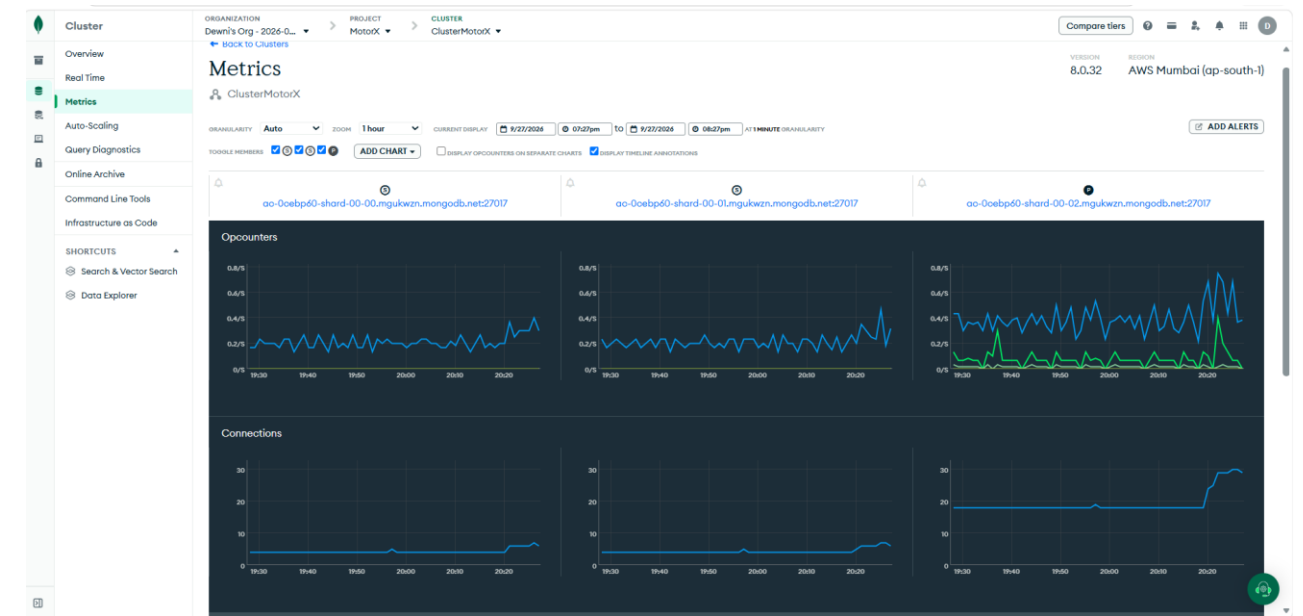


Figure 11: Atlas cluster metrics for ClusterMotorX (MongoDB 8.0.32, AWS Mumbai), 19:27–20:27 on 27 September 2026: operations per second and connections on the three replica-set members

Over that hour the three members of the replica set (two secondaries and the primary) handled well under one operation per second each, and the primary held about 20 connections, rising to about 30 near the end of the window as more services connected. The cluster is far from its limits at the current load. These figures are observations, not a load test: NF-03 remains the planned test of behaviour under many simultaneous users.

5.2 Reporting on Test Coverage

Coverage is reported in three ways: how many tests and cases ran and passed (Section 5.1 and Appendix D), which requirements have evidence (Appendix G), and how much of the code the tests executed. The third was measured with @vitest/coverage-v8, running each workspace’s full suite (npm run test:coverage): the backend and worker on 27 September 2026 in the isolated test stack, and the frontend on 28 September 2026 after its new component tests were added. Every test passed during the measurement.

Workspace	Statements	Branches	Functions	Lines
Backend	66.9% (1114/1665)	53.7% (582/1084)	63.4% (284/448)	72.6% (934/1287)
Worker	62.2% (448/720)	55.5% (221/398)	49.4% (88/178)	63.2% (350/554)
Frontend	69.0% (1410/2043)	68.4% (1193/1745)	63.4% (477/752)	71.2% (1144/1606)

Reading the figures: in the backend and worker the business rules are well exercised — buyer recommendations 98%, dealer module 88%, worker services, pipeline and jobs 82–90% of statements. The unexecuted code is mostly start-up and wiring (server.ts, worker.ts, configuration), the one-off migration command line and some repository helpers; these run in every deployment and are checked by the smoke runs rather than by unit tests. The frontend rose from 38% to 69% of statements (39% to 68% of branches) after component tests were written for the pages that had none: the dealer listing form, CSV upload and dashboard, the admin user, listing, audit-log and system-health pages, sign-in, the notification centre, the marketplace, and every API service module. What remains untested in the frontend is mainly the application shell and routing (App.tsx), the photo cropper, which draws on a canvas that the test browser cannot provide, and parts of registration and the sign-in session provider; these are exercised by the browser journeys instead. No coverage threshold is enforced; the figures are the baseline for the next cycle.

Report	Produced from	Refreshed	Read by
Automated results	Vitest JSON output of the three workspaces	On every push by CI; locally before each regeneration of this report	The developer who made the change
Case and test registers	The report generator (kept outside the repository)	Whenever any result or case changes	Whole team; mentor

Report	Produced from	Refreshed	Read by
Manual journey results	manual-results.csv, screenshots, Lighthouse reports	At the end of each browser test session	Team; merged into this report
Requirement traceability (Appendix G)	The generator, from the case list	With every regeneration	Mentor and examiners
Security scans	Gitleaks, Trivy and npm audit in CI	Every push, and a weekly scheduled run	The member who leads security
Deployment evidence	Smoke script against each stack; compose port check	After every deployment	Whoever deployed
Code coverage	@vitest/coverage-v8 (npm run test:coverage in each workspace)	Measured 27 Sep 2026; to be repeated before each release	Developers

Each report above feeds the generic test report below, which gives for every test field the date it was run, who ran it, how many tests or cases were executed, and how many passed and failed. Percentages are of the executed items; items not yet run are named in the comments rather than counted as failures.

Date	Tester	Test field	Executed	Pass	Fail	Pass %	Fail %	Comments
27 Sep 2026	CI and developers (automated)	Unit and component tests	291	291	0	100.0%	0.0%	All executed
27 Sep 2026	CI and developers (automated)	Integration tests — data (IT-01–IT-07, IT-11)	24	24	0	100.0%	0.0%	All executed
27 Sep 2026	CI and developers (automated)	Integration tests — HTTP journeys (IT-08–IT-10)	32	32	0	100.0%	0.0%	All executed
—	Group 23 testers	Integration cases — external services (IT-12–IT-15)	0	0	0	—	—	4 not yet run
27 Sep 2026	Group 23 testers	Contract and consistency checks (CT-01–CT-07)	6	5	1	83.3%	16.7%	1 not yet run; 1 failed (see Appendix F)
27 Sep 2026	Group 23 testers	End-to-end — system procedures (E2E-A1–A3)	3	1	2	33.3%	66.7%	All executed; 2 failed (see Appendix F)
—	Group 23 testers	End-to-end — browser journeys (E2E-01–E2E-15)	0	0	0	—	—	15 not yet run
27 Sep 2026	Group 23 testers	Performance (NF-01, NF-02)	1	1	0	100.0%	0.0%	1 not yet run
—	Group 23 testers	Load (NF-03)	0	0	0	—	—	1 not yet run
27 Sep 2026	Group 23 testers	Security and supply chain (NF-04, NF-05)	2	2	0	100.0%	0.0%	All executed
—	Group 23 testers	Search quality (NF-06)	0	0	0	—	—	1 not yet run
—	—	Total	359	356	3	99.2%	0.8%	23 not yet run

6. Risks, Dependencies, Assumptions, and Constraints

ID	Risk	Likelihood / impact	Mitigation	Contingency
R-01	The laptops and the EC2 stack share one Atlas database, so a data-changing test or migration on either changes the demo data.	Observed / High	Data-changing tests only on the motorx-test stack; migrations dry-run first; demo data created by test accounts.	Give EC2 its own Atlas database.
R-02	Browser journeys not all executed before submission.	Medium / High	Journeys scripted step by step and split between members (Section 8).	Report unexecuted journeys as such, alongside the API-level evidence.
R-03	Photos for 22 listings exist only in the EC2 MinIO volume.	Medium / High	Never run docker compose down -v on EC2; back up the volume.	Re-upload photos through the dealer portal.
R-04	The EC2 instance is too small to build all images at once (ETXTBSY, memory).	Observed / Medium	Build one service at a time with swap enabled.	Build images in CI and pull them on EC2.
R-05	The EC2 frontend ran the Vite development server rather than a production build.	Closed (28 Sep)	compose.ec2.yml now builds the production image served by nginx with compression and caching; Lighthouse Performance rose from 26 to 84 (Section 5.1.4).	Revert to the previous image if the production build fails on EC2.
R-06	External services unavailable during testing or the demo (Atlas DNS, Firebase, Hugging Face, Gmail SMTP).	Observed / High	Readiness endpoint; outbox retries; lexical search fallback; automated tests do not depend on them.	Show recorded evidence; repeat live checks when service returns.
R-07	API and worker Mongoose models of shared collections drift apart (CT-06 not automated).	Medium / High	Review both models in every change touching a shared collection.	Add an automated model-comparison check.
R-08	Sinhala and Tamil text has not been reviewed by native speakers.	Medium / Medium	Native readers in E2E-13.	Fall back to English for any text they reject.
R-09	Administrator accounts have no multi-factor sign-in.	Accepted / High	Strong passwords, few admin accounts, audit log of every admin action.	Enable Firebase multi-factor authentication.
R-10	Secrets exposed through shared .env files or screenshots.	Low / High	Gitleaks in CI; .env never committed; screenshots checked before sharing.	Rotate the affected keys immediately.
R-11	Burstable EC2 CPU credits distort performance and load results.	Medium / Medium	Record instance type and credit balance with each run.	Repeat runs on a fixed-performance instance.

Dependencies: Docker Desktop on the test laptop; the locked npm dependency set; internet access to Atlas, Firebase and Hugging Face for live checks; the EC2 instance and its Elastic IP; members' time for browser journeys.

Assumptions: the team keeps the module ownership in team-work-plan.md; the Firebase project used for browser journeys holds only test accounts; the demonstration data on Atlas was created by the team and contains no real customer records.

Constraints: three members who are also the developers; a laptop whose memory limits parallel test runs; one EC2 instance serving as both staging and demonstration; no browser automation installed in this cycle, so browser journeys are manual.

7. References

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- [17] Supplied template “6 Template for Test plan.docx” (Rational Unified Process master test plan).

8. Responsibilities, Staffing and Training Needs

Every member of Group 23 both builds and tests MotorX. The roles below follow the module ownership in team-work-plan.md: Silva T.D.R. (230616B) owns authentication, dealers and the marketplace; Somarathna M.D.A.M. (230621K) owns inventory and the ETL worker; Thadshakan J. (230633A) owns search, notifications and administration. The member who writes a change also writes its automated tests; the lead of a test type is responsible for its evidence being complete and current.

8.1 Ownership by Test Type

Test type	Lead	Supporting	Duties
3.3.1 Data integrity	Somarathna	Silva	Keep the replica-set suites and the drill working; add a data test for every new unique rule or shared collection; run CT-06 reviews.
3.3.2 Function	Module owner	All	Tests for every requirement in the module; keep Appendix G current.
3.3.3 User interface	Silva	Thadshakan	Component tests for new pages; run E2E-12 and E2E-15; organise the native-reader review (E2E-13).
3.3.4 Performance profiling	Somarathna	Thadshakan	Rerun the benchmark after pipeline changes; collect NF-02 percentiles on EC2.
3.3.5 Load	Thadshakan	Somarathna	Write and run the k6 scenario (NF-03) in an agreed quiet window.
3.3.6 Security	Silva	All	Access-control journeys, CI scan triage, security-group and port reviews after each deployment.
3.3.7 Failover and recovery	Somarathna	Silva	Drill reruns before submission; recovery tests for every new background job.
3.3.8 Configuration	Thadshakan	Silva	Smoke runs and port checks after every deployment; the browser matrix during journeys.

8.2 Shared Duties

- Whoever fixes a finding adds the test that guards it and updates Appendix F.
- Whoever deploys runs the smoke script and CT-04 afterwards and saves the output as evidence.
- Each member reviews test evidence for a module they did not write before it is counted as passed.
- The generator is rerun after every session so the report and the registers never disagree.

8.3 Skills to Build

Skill	Needed for	Current level	How it will be learned
Vitest, Testing Library, supertest	All automated levels	In daily use	Existing tests serve as examples; reviewed in pull requests.
MongoDB replica sets, indexes, transactions	3.3.1	Held mainly by Somarathna	One member pairs with Somarathna to rerun the data suites and the drill unaided.
Playwright	Automating Appendix C	Not yet used	Automate E2E-07 first as a reference, then the other journeys.
k6 load scripting	3.3.5	Not yet used	A 5-user trial against EC2 to calibrate before the full run.
Lighthouse and WCAG checks	3.3.3	Basic	Run Lighthouse on three pages and fix what it reports.
Operating Docker on EC2 (swap, disk, sequential builds, logs)	3.3.8; deployments	Learned during this cycle	A short runbook written from the 27 Sep deployment.

Skill	Needed for	Current level	How it will be learned
Reviewing Sinhala and Tamil copy	3.3.3	Through members' own networks	Brief native readers with the E2E-13 script.

8.4 Staffing Limits

- Roles overlap because the group has three members; each test type still has one lead so someone is always accountable.
- Load runs and drills need the relevant stack to themselves and are booked in advance.
- Only members with EC2 and Atlas administrator access may deploy or run migrations.
- Browser journeys take roughly a day of combined effort, the largest remaining time cost of this plan.

9. Test Environment Specification

Environment	Used for	Make-up	Limits
Developer laptop (compose project motorx)	Everyday development; frontend tests; E2E-A2 smoke.	Windows 11; Docker Desktop; backend, worker, frontend, Redis and MinIO containers; Atlas over the internet.	Shares the Atlas database with EC2 (R-01); limited memory, so frontend tests run two workers.
Isolated test stack (compose project motorx-test)	All backend and worker suites; benchmark; crash drill.	mongo:7 as replica set rs0 (database motorx_test), redis:7, MinIO with a test bucket; Node 24 containers built from the development stage.	Disposable; not representative of internet latency.
GitHub Actions CI	Build, all three test suites, Gitleaks, Trivy and npm audit on every push; weekly scheduled run.	Ubuntu runners; Node 24; MongoDB replica set service.	Placeholder credentials only (F-11); one rate-limit test needing real Redis is skipped.
EC2 demonstration deployment	E2E-A3 smoke; browser journeys; NF-02 and NF-03.	Ubuntu 26.04 in ap-south-1; Elastic IP 13.207.143.45; Docker Compose with compose.ec2.yml; Node 24; MinIO bucket motorx-staging; shared Atlas database.	Single instance; Vite development server for the frontend (R-05); only ports 22, 80, 443, 3000 and 4173 open.
Amazon ECS (planned)	Future production.	Defined in .github/workflows/deploy.yml.	Not provisioned in this cycle.

9.1 Differences Between Environments

- HTTP journeys replace Firebase token checks and the S3 client; only the smoke runs and browser journeys use the real services.
- The test stack uses a local replica set, while the laptops and EC2 use Atlas; latency and index build times differ.
- EC2 serves the frontend through the Vite development server; a production image would serve a static build.
- MinIO stands in for Amazon S3 on every stack; S3-specific behaviour such as bucket policies is not tested.
- CI has no real Firebase credentials, so a test that loads the real Firebase setup fails there and nowhere else.

10. Entry, Exit and Suspension Criteria

The following criteria define when testing can start, when a level is complete, and when testing must pause. The status columns report the available evidence; preparing this plan does not imply that every execution criterion was met.

10.1 Entry Criteria

Level	May start when
Unit and component	The workspace builds and the behaviour has an agreed expected result (an SRS clause or a documented extension).
Integration	Unit tests of the module pass; the motorx-test stack is healthy with an elected replica-set primary; the database guard confirms the target.
Contract and consistency	Both sides of the agreement exist at the same commit (for example the new Compose file and the service it configures).
End-to-end (system)	Integration tests pass; the target stack is deployed from a known commit and its health endpoints answer.
End-to-end (browser)	The API-level journey for the same flow passes; test accounts exist in the test Firebase project; the tester has the journey script.
Performance and load	The flows under measurement are functionally correct; the stack is reserved; instance type and data volume are recorded.

10.2 Exit Criteria

Level	Complete when	Status this cycle
Unit and component	Every module in scope has tests including invalid-input cases, and all pass on two consecutive runs.	Met
Integration	Every database-enforced rule has a real-database test; all journeys pass.	Met for automated cases; IT-12–IT-15 open
Contract and consistency	All seven checks executed and passed.	Not met: CT-06 not run, CT-07 failed
End-to-end	System procedures pass on the deployed stack; all High-priority browser journeys executed and passed.	Not met
Performance and load	NF-01 to NF-03 executed and within their targets, or reported as unmet.	Partly met: NF-01 only

10.3 Submission Exit Criteria

Criterion	Status
All automated suites pass in CI with no required check bypassed.	Met (348/348; CI green on PR #17 after F-11)
No open Critical or High finding without an accepted workaround.	Not met: F-01 partly fixed
High-priority browser journeys executed and passed on the EC2 deployment.	0 of 13 passed
PSR-05 measured and met; PSR-01–03 measured or reported as unmet.	PSR-05 met; others outstanding
Smoke run passes on the deployment used for the demonstration.	Not met: 2 checks failing (F-01)
Every SRS area in Appendix G has executed evidence or a stated gap.	Met

10.4 Suspension and Resumption

Suspend testing when	Resume when
----------------------	-------------

Suspend testing when	Resume when
The database guard refuses the target, or a test is found pointing at Atlas or the EC2 data.	The target is corrected and verified, and any affected data is checked against its expected state.
Atlas, Firebase or the network is unavailable.	Readiness returns 200 and the smoke run passes again; interrupted cases are rerun from the start.
A deployment is half-finished (for example an image build failed on EC2).	All containers report healthy on the intended commit.
A shared collection's shape changed in one process but not the other.	Both models agree (CT-06 review) and the data suites pass.
Failures keep repeating one known cause.	The cause is fixed and the affected level is re-entered from its entry criteria.

11. Test Schedule and Phasing

The planned submission deadline was 27 September 2026. Phases are shown in sequence without individual dates. The scope of each phase was defined for submission; execution results and outstanding findings are recorded separately in Section 5 and the appendices.

Phase	Defined scope	Completion requirement
1. Baseline	Prepare the test plan and case register; establish unit and component coverage.	Plan, register and initial test results recorded.
2. Hardening	Cover Node 24, security scanning, safe file processing, leases, checkpoints, reaper and outbox.	Automated checks and regression evidence recorded.
3. Journeys and evidence	Cover HTTP journeys, frontend components, the import benchmark, crash recovery and local smoke checks.	Results and evidence linked to the relevant cases.
4. Deployment	Verify the EC2 runtime, ports, allowed hosts, deployment smoke checks and code coverage.	Deployment checks and findings documented.
5. Browser journeys	Cover E2E-01 to E2E-15 on EC2 and photo re-upload verification for F-01.	Each case assigned a result or an explicit execution gap.
6. Measurement	Cover NF-02 response-time percentiles, NF-03 load and Lighthouse audits.	Measurements recorded; unmeasured targets identified.
7. Submission and sign-off	Generate the report and registers; provide Appendices F and G for mentor review.	Report prepared with findings, coverage gaps and approval status stated.

11.1 Activities Across Phases

- Retain automated suites and security scans in CI throughout subsequent phases.
- Classify findings using Section 12 and record their resolution or acceptance status.
- Regenerate the report and registers from the available evidence for submission.

12. Defect Severity and Priority Definitions

Each finding in Appendix F is given two ratings. Severity describes how badly the problem affects users or their data. Priority defines the required resolution stage: demonstration, submission, next iteration or backlog. The two are rated separately on purpose: a serious fault in a feature nobody uses yet can wait, while a small flaw on the first page every visitor sees may need fixing straight away.

12.1 Severity

Severity	What it means	Examples from this cycle
Critical	Data is lost or silently corrupted, or one user can see or change another user's data.	None found. A dealer reading another dealer's listings, or an import creating duplicate vehicles, would be Critical.
High	A main feature does not work, or a security weakness is only partly covered.	F-01 some listing photos not loading; F-09 re-attached photos being ignored; F-14 Redis and MinIO reachable from outside the server; F-15 the deployed site served without HTTPS.
Medium	A secondary feature does not work, or a main feature fails only in unusual conditions and a workaround exists.	F-11 a test that failed only in CI; F-12 the server refusing its public host name; F-13 image builds failing on EC2; F-18 two dependency advisories on code paths MotorX uses.
Low	Appearance, wording or speed issues that do not stop anything working.	F-03 older photos without a small copy, so pages load more slowly on phones; F-16 a raw status name shown to dealers.

12.2 Priority

Priority	What it means
Before demo	Fixed before the deployed system is next demonstrated.
Before submission	Fixed before the final submission, or accepted with the reason written down.
Next iteration	Planned for the first development iteration after submission.
Backlog	Recorded, and fixed the next time that part of the system is changed, or accepted.

12.3 Recording and Lifecycle

For every finding we record an ID, a short description, its severity and priority, the test or activity that found it, the environment it was found in, its current status, and the test or check that proves it is fixed. A finding is Open until the fix is merged, when it becomes Fixed. It is Closed only after the proving test passes in CI, or the proving check has been repeated on the environment where the problem appeared. When a finding will not be fixed, it is marked Accepted, together with the reason and the name of the person who accepted it.

13. Test Data Policy

This policy defines the permitted data sources, isolation requirements and cleanup rules for every test environment.

No real customer data is used in any test stack or committed to the repository. Test data is either generated by the tests themselves or created by team members through the application using test accounts. This matters because MotorX stores identity documents and business registration certificates, and because the laptops and the EC2 stack share one Atlas database.

13.1 Rules

- Automated tests build their own data in memory — plates such as CAX-1001, vehicles, users, CSV rows, images made with sharp and hand-built PDFs — and remove it afterwards.
- Automated tests may only use a database named `motorx_test` on a local host; the guard stops any other target before connecting.
- Demonstration data on Atlas is entered by team members through the app, using test accounts and invented business details.
- Uploaded photos are rebuilt without metadata, so location data from a phone camera never reaches storage.
- E-mail from test runs goes only to team inboxes.
- Keys and passwords never appear in fixtures, evidence files or screenshots; Gitleaks scans every push.

13.2 Test Data in Each Environment

Environment	Where the data comes from	Size
Isolated test stack	Created by each test and cleared by the next; the benchmark and the drill generate their own CSV files.	From a few documents up to 60,000 rows (drill).
CI	As the test stack, recreated on every run.	Small.
Laptops and EC2 (shared Atlas)	Team-created demo accounts, dealers and listings, plus CSV imports of the sample file <code>source-csv.txt</code> .	Tens of active listings.
Browser journeys	Fresh test accounts per journey in the test Firebase project.	A handful per journey.

13.3 Invalid and Hostile Inputs

Several cases deliberately use bad input: CSV rows with missing or mistyped fields, category mismatches and duplicate plates; files that only start like a JPEG; a 41-megapixel image that is tiny on disk; photos carrying GPS EXIF data; PDFs containing JavaScript, launch actions or embedded files; a program renamed with a `.pdf` extension; ZIPs with too many entries, oversized entries, backslash paths, wrapper folders or no folders at all. These inputs are created inside the tests that need them and are never stored in shared environments.

Appendix A – Unit and Component Test Cases

48 case groups, one per test file, 291 tests in total. Individual test names and durations are in automated-test-register.csv.

UT-01 — Test database target safety

Field	Specification and result
Requirement / priority	Test infrastructure; RR-09 / High
Set-up and data	Local, Docker, non-test, remote and wrong-scheme URIs; missing TEST_MONGODB_URI.
Steps	Validate each URI before any connection.
Expected result	Only a motorx_test database on an approved host is accepted; never falls back to MONGODB_URI.
Execution	Executed 26 Sep 2026: 8/8 passed
Evidence	apps/backend/src/test/db.safety.test.ts; test-evidence/backend-results.json

UT-02 — Production database URI guard

Field	Specification and result
Requirement / priority	PSR-10; RR-09 / High
Set-up and data	Atlas SRV, TLS/non-TLS, local and test database names.
Steps	Check each URI with findProductionMongoUriProblems.
Expected result	Production refuses unencrypted, local, test/dev databases and URIs without a database name.
Execution	Executed 26 Sep 2026: 10/10 passed
Evidence	apps/backend/src/config/mongoUri.test.ts; test-evidence/backend-results.json

UT-03 — Admin request validation

Field	Specification and result
Requirement / priority	FR-ADMIN-02-05 / High
Set-up and data	Default and bounded pages; user/listing filters; invalid status.
Steps	Parse query schemas.
Expected result	Defaults and valid filters parse; unsupported values fail.
Execution	Executed 26 Sep 2026: 4/4 passed
Evidence	apps/backend/src/modules/admin/admin.validation.test.ts; test-evidence/backend-results.json

UT-04 — Dealer application validation

Field	Specification and result
Requirement / priority	FR-DEALER-02,09-13 / High
Set-up and data	Multipart fields; incomplete application; missing rejection reason.
Steps	Parse application and review schemas.
Expected result	Form values normalized; incomplete data and missing rejection reason rejected.
Execution	Executed 26 Sep 2026: 3/3 passed
Evidence	apps/backend/src/modules/dealers/dealer.validation.test.ts; test-evidence/backend-results.json

UT-05 — Dealer document content checks

Field	Specification and result
Requirement / priority	FR-DEALER-09; PSR-13–14 / High
Set-up and data	Plain PDF; PDFs with JavaScript, launch action, embedded file; renamed executable; images with metadata; fake JPEG.
Steps	Detect type from bytes and sanitize each file.
Expected result	Only safe PDFs and re-encoded images are kept; dangerous or disguised files are rejected.
Execution	Executed 26 Sep 2026: 9/9 passed
Evidence	apps/backend/src/modules/dealers/dealerDocument.content.test.ts; test-evidence/backend-results.json

UT-06 — CSV upload validation

Field	Specification and result
Requirement / priority	FR-UPLOAD-01–03,07 / High
Set-up and data	Car and motorcycle headers; missing headers; binary files; paging.
Steps	Validate files, headers and pagination.
Expected result	Matching headers accepted per category; bad content and missing fields rejected; paging bounded.
Execution	Executed 26 Sep 2026: 6/6 passed
Evidence	apps/backend/src/modules/inventory/inventory.validation.test.ts; test-evidence/backend-results.json

UT-07 — Upload acceptance and controlled retry

Field	Specification and result
Requirement / priority	FR-UPLOAD-04–06; RR-06–08 / High
Set-up and data	Queue available, down, and hanging; dealer at listing limit; job-creation failure; failed and non-failed uploads.
Steps	Accept uploads and request retries with mocked queue and storage.
Expected result	Uploads are kept for later queuing when Redis is down, the dealer is never kept waiting, storage is cleaned on failure, and only the owner's failed jobs can be retried.
Execution	Executed 26 Sep 2026: 8/8 passed
Evidence	apps/backend/src/modules/inventory/inventory.service.test.ts; test-evidence/backend-results.json

UT-08 — Listing validation and identity

Field	Specification and result
Requirement / priority	FR-MARKET-02–05; FR-ETL-19–21 / High
Set-up and data	Plate variants, pagination, inverted ranges, empty edits, image order, category/powertrain fixtures.
Steps	Normalize registrations and parse listing schemas.
Expected result	Equivalent plates share one identity; invalid edits, ranges and attributes fail.
Execution	Executed 26 Sep 2026: 12/12 passed
Evidence	apps/backend/src/modules/marketplace/listing.validation.test.ts; test-evidence/backend-results.json

UT-09 — Image signature check

Field	Specification and result
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Field	Specification and result
Requirement / priority	FR-MARKET-13; PSR-14 / High
Set-up and data	JPEG, PNG, WebP headers; spoofed JPEG; GIF.
Steps	Check each file signature.
Expected result	Supported signatures accepted; spoofed and unsupported files rejected.
Execution	Executed 26 Sep 2026: 2/2 passed
Evidence	apps/backend/src/modules/marketplace/listingImage.service.test.ts; test-evidence/backend-results.json

UT-10 — Image re-encoding

Field	Specification and result
Requirement / priority	FR-MARKET-13; PSR-13–14 / High
Set-up and data	JPEG with GPS EXIF and rotation; trailing hidden data; 41-megapixel bomb; fake image; SVG/GIF; PNG document.
Steps	Re-encode each image.
Expected result	Output is a clean WebP (or PNG for documents), upright, without metadata or hidden data; bombs and disallowed formats refused.
Execution	Executed 26 Sep 2026: 7/7 passed
Evidence	apps/backend/src/shared/utils/imageReencode.test.ts; test-evidence/backend-results.json

UT-11 — Photo migration and small copies

Field	Specification and result
Requirement / priority	PSR-13; Extension (small photo copies) / High
Set-up and data	Old root-level photos with GPS; already migrated photos; dry run; CDN URL; photo without small copy; missing and undecodable objects.
Steps	Run the migration against an in-memory bucket.
Expected result	Photos moved as clean WebP; re-runs skip finished work; dry run changes nothing; an 800 px small copy is created once and its URL recorded; problems reported, not hidden.
Execution	Executed 26 Sep 2026: 6/6 passed
Evidence	apps/backend/src/scripts/listingImageMigration.test.ts; test-evidence/backend-results.json

UT-12 — Search query analysis

Field	Specification and result
Requirement / priority	FR-SEARCH-08–20; FR-ETL-28 / High
Set-up and data	“automatic SUV under 2 million near Colombo”; “toyota corola”; mileage vs price; oversized query; vector fixture.
Steps	Analyze queries and normalize embeddings.
Expected result	Correct structured filters and typo corrections; bounded input; normalized 384-length vectors.
Execution	Executed 26 Sep 2026: 5/5 passed
Evidence	apps/backend/src/modules/search/search.queryAnalyzer.test.ts; test-evidence/backend-results.json

UT-13 — Vehicle similarity scoring

Field	Specification and result
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Field	Specification and result
Requirement / priority	Extension (similar vehicles, recommendations) / High
Set-up and data	Same model, same make, other make; price and year differences; case/spacing; view order.
Steps	Score candidates against one or several viewed vehicles.
Expected result	Closer vehicles score higher; recent views weigh more; ties keep newest-first order.
Execution	Executed 26 Sep 2026: 5/5 passed
Evidence	apps/backend/src/modules/buyers/buyer.similarity.test.ts; test-evidence/backend-results.json

UT-14 — Pagination metadata

Field	Specification and result
Requirement / priority	FR-SEARCH-06 / High
Set-up and data	Partial final page; empty collection.
Steps	Build pagination metadata.
Expected result	Correct page counts; zero pages when empty.
Execution	Executed 26 Sep 2026: 2/2 passed
Evidence	apps/backend/src/shared/utils/pagination.test.ts; test-evidence/backend-results.json

UT-15 — Token verification cache

Field	Specification and result
Requirement / priority	FR-USER-07-09; PSR-08 / High
Set-up and data	Valid, revoked, expiring and missing tokens.
Steps	Call the middleware repeatedly with a mocked Firebase Admin SDK.
Expected result	Firebase (with revocation) checked once per cache period; never trusted past token expiry; failures not cached.
Execution	Executed 26 Sep 2026: 4/4 passed
Evidence	apps/backend/src/shared/middleware/verifyFirebaseToken.test.ts; test-evidence/backend-results.json

UT-16 — CSV extraction

Field	Specification and result
Requirement / priority	FR-ETL-05-06 / High
Set-up and data	Multi-batch CSV stream; wrong column count.
Steps	Stream records and record progress.
Expected result	Bounded batches with cumulative progress; malformed rows fail.
Execution	Executed 26 Sep 2026: 2/2 passed
Evidence	apps/worker/src/pipeline/extract.test.ts; test-evidence/worker-results.json

UT-17 — Vehicle normalization

Field	Specification and result
Requirement / priority	FR-ETL-07,23 / High
Set-up and data	Whitespace, enum aliases, price suffixes, blank optional cells, motorcycle rows, titles cut short to a prefix of make and model.

Field	Specification and result
Steps	Normalize rows per category.
Expected result	Consistent values; blanks stay unset; category-specific attributes; a truncated title such as “Toyota Pre” becomes “Toyota Premio”, while a longer title is kept.
Execution	Executed 26 Sep 2026: 6/6 passed
Evidence	apps/worker/src/pipeline/normalize.test.ts; test-evidence/worker-results.json

UT-18 — Batch transformation

Field	Specification and result
Requirement / priority	FR-ETL-08–17; RR-05 / High
Set-up and data	Mixed valid/invalid rows; electric car row.
Steps	Prepare a batch with known starting row numbers.
Expected result	Valid rows kept; invalid rows isolated with their CSV row numbers.
Execution	Executed 26 Sep 2026: 3/3 passed
Evidence	apps/worker/src/pipeline/transform.test.ts; test-evidence/worker-results.json

UT-19 — Category and powertrain rules

Field	Specification and result
Requirement / priority	FR-MARKET-03; FR-ETL-08–11 / High
Set-up and data	Petrol, diesel, hybrid, electric, plug-in hybrid cars and five other categories.
Steps	Validate valid and invalid combinations.
Expected result	Engine or battery data required as appropriate; field-level errors returned.
Execution	Executed 26 Sep 2026: 17/17 passed
Evidence	apps/worker/src/pipeline/validate.test.ts; test-evidence/worker-results.json

UT-20 — CSV ETL orchestration

Field	Specification and result
Requirement / priority	FR-ETL-13–22,31–33; RR-06–08 / High
Set-up and data	Mocked storage/repositories: mixed rows, malformed CSV, duplicates, temporary storage failure, exhausted attempts, crash mid-import, listing limit, lost lease.
Steps	Run the import service and inspect writes and counters.
Expected result	Accurate counters; permanent vs temporary failures handled differently; resumes from the last checkpoint with no row imported twice; stops when the lease is lost.
Execution	Executed 26 Sep 2026: 9/9 passed
Evidence	apps/worker/src/services/uploadJob.service.test.ts; test-evidence/worker-results.json

UT-21 — ZIP photo processing

Field	Specification and result
Requirement / priority	FR-MARKET-12–16; RR-06; Extension (small copies) / High
Set-up and data	ZIPs with slash/backslash paths, an extra wrapper folder, photos whose extension does not match their content, unknown folders, root files, oversize entries, fake and huge images, storage errors, retries, and a ZIP with no photo in any folder.

Field	Specification and result
Steps	Process ZIPs with mocked storage and repositories.
Expected result	Photos matched by the folder that contains them (normalized plate), cleaned and resized, an 800 px copy stored and linked; a ZIP with no photos in folders fails at once with instructions; unsafe archives fail at once; retries never attach a photo twice.
Execution	Executed 26 Sep 2026: 17/17 passed
Evidence	apps/worker/src/services/imageProcessing.service.test.ts; test-evidence/worker-results.json

UT-22 — Job lease renewal

Field	Specification and result
Requirement / priority	RR-06-08 / High
Set-up and data	Long-running job; takeover by another worker; renewal error; job end.
Steps	Hold a lease with fake timers.
Expected result	Lease renewed every third of its duration; loss detected; temporary errors tolerated; renewal stops at the end.
Execution	Executed 26 Sep 2026: 4/4 passed
Evidence	apps/worker/src/services/jobLease.test.ts; test-evidence/worker-results.json

UT-23 — Retry classification

Field	Specification and result
Requirement / priority	RR-06-08 / High
Set-up and data	Network, throttling, 5xx, MongoDB unreachable; missing file, access denied, parse and duplicate errors.
Steps	Classify each error.
Expected result	Only temporary failures are retried.
Execution	Executed 26 Sep 2026: 12/12 passed
Evidence	apps/worker/src/services/transientError.test.ts; test-evidence/worker-results.json

UT-24 — Lost-job reconciliation

Field	Specification and result
Requirement / priority	FR-ETL-03; RR-06-08 / High
Set-up and data	Pending uploads with missing, waiting, delayed or active queue messages; exhausted budgets.
Steps	Run one reaper cycle with mocked queue.
Expected result	Lost jobs re-queued under the right job ID; jobs with a live message untouched; exhausted jobs failed instead of looping.
Execution	Executed 26 Sep 2026: 6/6 passed
Evidence	apps/worker/src/jobs/reaper.job.test.ts; test-evidence/worker-results.json

UT-25 — Email outbox delivery

Field	Specification and result
Requirement / priority	FR-NOTIFY-06-07; RR-08 / High
Set-up and data	Due emails; SMTP failures on attempts 1–5; deleted recipient.

Field	Specification and result
Steps	Run outbox cycles with a mocked mailer.
Expected result	Each email sent once; retries after 1 min, 5 min ... 2 h; failed after the fifth attempt; bounded work per cycle.
Execution	Executed 26 Sep 2026: 7/7 passed
Evidence	apps/worker/src/jobs/emailOutbox.job.test.ts; test-evidence/worker-results.json

UT-26 — Document retention

Field	Specification and result
Requirement / priority	FR-DEALER-13; PSR-15 / High
Set-up and data	Decisions older than 90 days; storage delete failure.
Steps	Run one retention cycle.
Expected result	Files deleted then record cleared; a failed delete is retried next cycle.
Execution	Executed 26 Sep 2026: 3/3 passed
Evidence	apps/worker/src/jobs/documentRetention.job.test.ts; test-evidence/worker-results.json

UT-27 — Stale-stock reminders

Field	Specification and result
Requirement / priority	Extension (stale stock) / High
Set-up and data	Dealers with stale listings; claim won or lost; one dealer failing.
Steps	Run one reminder cycle with mocked repositories.
Expected result	60-day cutoff and 7-day repeat applied; each dealer gets their own count once; one failure does not stop the others.
Execution	Executed 26 Sep 2026: 3/3 passed
Evidence	apps/worker/src/jobs/staleListingReminder.job.test.ts; test-evidence/worker-results.json

UT-28 — Role-protected pages

Field	Specification and result
Requirement / priority	FR-USER-10–12 / High
Set-up and data	Signed-out, allowed, disallowed and pending-applicant users.
Steps	Render protected routes.
Expected result	Redirect to login, show page, show “Access Restricted” or send to the application status page as appropriate.
Execution	Executed 27 Sep 2026: 4/4 passed
Evidence	apps/frontend/src/features/auth/components/RoleGuard.test.tsx; test-evidence/frontend-results.json

UT-29 — Email verification banner

Field	Specification and result
Requirement / priority	FR-USER-04 / High
Set-up and data	Verified, unverified and signed-out users.
Steps	Render the banner and resend/refresh.
Expected result	Shown only when needed; resend works; hides once verified.

Field	Specification and result
Execution	Executed 27 Sep 2026: 4/4 passed
Evidence	apps/frontend/src/features/auth/components/EmailVerificationBanner.test.tsx; test-evidence/frontend-results.json

UT-30 — Per-account data isolation

Field	Specification and result
Requirement / priority	FR-USER-12; PSR-09 / High
Set-up and data	Sign-out; a different account signing in on the same browser.
Steps	Switch accounts and inspect the query cache.
Expected result	Previous account's cached data is cleared before the next user can see it.
Execution	Executed 27 Sep 2026: 2/2 passed
Evidence	apps/frontend/src/features/auth/context/AuthProvider.test.tsx; test-evidence/frontend-results.json

UT-31 — Dealer application from an existing account

Field	Specification and result
Requirement / priority	FR-DEALER-09–13 / High
Set-up and data	Rejected and pending applications for a signed-in buyer.
Steps	Render the application page and resubmit.
Expected result	Rejection reason shown, answers pre-filled, no password asked, resubmission sent with new documents; pending applicants redirected.
Execution	Executed 27 Sep 2026: 3/3 passed
Evidence	apps/frontend/src/features/auth/pages/RegisterPage.test.tsx; test-evidence/frontend-results.json

UT-32 — Dealer profile editing

Field	Specification and result
Requirement / priority	FR-DEALER-01–04 / High
Set-up and data	Approved dealer profile; server error.
Steps	Edit and save.
Expected result	Verified name and registration read-only; edits saved and confirmed; server errors shown.
Execution	Executed 27 Sep 2026: 3/3 passed
Evidence	apps/frontend/src/portals/dealer/pages/DealerProfile.test.tsx; test-evidence/frontend-results.json

UT-33 — Upload details and publish-all

Field	Specification and result
Requirement / priority	FR-UPLOAD-06–08; Extension (bulk) / High
Set-up and data	Failed CSV, failed photos, refused retry, successful upload with 9 drafts.
Steps	Render the page and use Retry and Publish all.
Expected result	Failure reasons and retries work; all drafts of the upload are published in one request after confirmation.
Execution	Executed 27 Sep 2026: 5/5 passed
Evidence	apps/frontend/src/portals/dealer/pages/UploadDetails.test.tsx; test-evidence/frontend-results.json

UT-34 — Bulk and stale-stock tools

Field	Specification and result
Requirement / priority	FR-DEALER-05–08; Extension (bulk, stale) / High
Set-up and data	Stats with 2 stale listings; draft and stale listings.
Steps	Use the banner, select-all bulk publish, one-click “Still available” and the price dialog.
Expected result	Stale banner opens the stale list; one request publishes all selected; age shown; price cut previewed (5,400,000 at 10%) then applied; skipped listings explained.
Execution	Executed 27 Sep 2026: 5/5 passed
Evidence	apps/frontend/src/portals/dealer/pages/ListingManager.test.tsx; test-evidence/frontend-results.json

UT-35 — Admin approvals, dashboard and monitoring

Field	Specification and result
Requirement / priority	FR-ADMIN-03–08 / High
Set-up and data	Deep links with status/applicationId/uploadId; failing sections; date ranges.
Steps	Render admin pages from links.
Expected result	Correct tab and exact application highlighted; failures shown as unavailable, not zero; filters passed to the server.
Execution	Executed 27 Sep 2026: 6/6 passed
Evidence	apps/frontend/src/portals/admin/pages/adminPages.test.tsx; test-evidence/frontend-results.json

UT-36 — Vehicle page on phones

Field	Specification and result
Requirement / priority	FR-MARKET-07–11; Extension (mobile, discovery) / High
Set-up and data	Listing with three photos (small copies) and a dealer phone 077 123 4567.
Steps	Render, swipe, tap arrows, scroll vertically.
Expected result	Call/WhatsApp/Email bar with the WhatsApp message ready (94771234567); small copy chosen via srcset; swipe changes photo, vertical scroll does not; vehicle remembered for recommendations; similar vehicles shown.
Execution	Executed 27 Sep 2026: 4/4 passed
Evidence	apps/frontend/src/portals/buyer/pages/VehicleDetails.test.tsx; test-evidence/frontend-results.json

UT-37 — Compare vehicles

Field	Specification and result
Requirement / priority	Extension (compare) / High
Set-up and data	Four vehicles; two vehicles with different price/year/mileage; one unavailable.
Steps	Add to compare, open comparison.
Expected result	At most three; full list explained; best price, year and mileage highlighted; unavailable vehicle stated.
Execution	Executed 27 Sep 2026: 4/4 passed
Evidence	apps/frontend/src/features/compare/compare.test.tsx; test-evidence/frontend-results.json

UT-38 — Mobile navigation and card tables

Field	Specification and result
Requirement / priority	UR-01–04; Extension (mobile) / High
Set-up and data	Portal with two pages; table with two columns.
Steps	Open menu, press Escape, choose a page; render a table.
Expected result	Drawer opens with focus inside; closes on Escape (focus returns) and after navigation; every cell labelled with its column.
Execution	Executed 27 Sep 2026: 3/3 passed
Evidence	apps/frontend/src/shared/components/mobileLayout.test.tsx; test-evidence/frontend-results.json

UT-39 — Sinhala and Tamil

Field	Specification and result
Requirement / priority	Extension (languages) / High
Set-up and data	All dictionaries; Tamil browser preference.
Steps	Compare dictionaries; switch language; reload.
Expected result	Every message translated with identical placeholders; whole site switches, page lang set, choice remembered; Tamil chosen automatically for a Tamil browser.
Execution	Executed 27 Sep 2026: 4/4 passed
Evidence	apps/frontend/src/shared/i18n/i18n.test.tsx; test-evidence/frontend-results.json

UT-40 — WhatsApp number formatting

Field	Specification and result
Requirement / priority	Extension (mobile contact) / High
Set-up and data	Local, +94, 9-digit, foreign and invalid numbers.
Steps	Convert numbers and build links.
Expected result	Sri Lankan numbers get 94; foreign numbers kept; undialable numbers give no link.
Execution	Executed 27 Sep 2026: 3/3 passed
Evidence	apps/frontend/src/shared/utls/phone.test.ts; test-evidence/frontend-results.json

UT-41 — Queue publishing for repeat photo uploads

Field	Specification and result
Requirement / priority	FR-UPLOAD-08; RR-06–08 / High
Set-up and data	Existing queue jobs that are completed, failed, waiting, active or delayed.
Steps	Publish CSV and photo jobs with a mocked queue.
Expected result	A finished job is removed and queued again under the same ID, so photos attached a second time are processed; a job still waiting or running is never duplicated (finding F-09).
Execution	Executed 26 Sep 2026: 4/4 passed
Evidence	apps/backend/src/modules/inventory/inventory.queue.test.ts; test-evidence/backend-results.json

UT-42 — Frontend API services

Field	Specification and result
Requirement / priority	FR-MARKET-01–16; FR-SEARCH-01–07; FR-UPLOAD-01–08; FR-DEALER-09–11; FR-ADMIN-02–10 / High

Field	Specification and result
Set-up and data	Stubbed HTTP client; listing, buyer, inventory, dealer and admin services.
Steps	Call every service function; inspect URLs, parameters, form data and returned models.
Expected result	Right endpoint and parameters for each call; server shapes converted to the UI model; invalid dealer replies rejected; a failed document download closes its tab.
Execution	Executed 27 Sep 2026: 13/13 passed
Evidence	apps/frontend/src/shared/services/apiServices.test.ts; test-evidence/frontend-results.json

UT-43 — Admin users, listings, audit log and system health pages

Field	Specification and result
Requirement / priority	FR-ADMIN-02-08; FR-USER-10 / High
Set-up and data	Mocked admin API with users, listings, audit events and health results, including failures.
Steps	Filter, suspend and reactivate users; archive listings; filter and page the audit log; open system health.
Expected result	Only the changed row updates; archiving asks first; empty, filtered-empty and failed states are told apart; each service status is shown.
Execution	Executed 27 Sep 2026: 9/9 passed
Evidence	apps/frontend/src/portals/admin/pages/adminManagement.test.tsx; test-evidence/frontend-results.json

UT-44 — Dealer listing form

Field	Specification and result
Requirement / priority	FR-MARKET-01-05,12-15 / High
Set-up and data	Mocked listing API; stand-in photo cropper; new and existing listings.
Steps	Create, edit and retry listings; switch categories and fuel types; crop, skip, reorder and remove photos.
Expected result	Only fields that apply are sent; electric vehicles need battery details; a failed photo upload keeps the saved listing and offers a retry.
Execution	Executed 27 Sep 2026: 8/8 passed
Evidence	apps/frontend/src/portals/dealer/pages/ListingForm.test.tsx; test-evidence/frontend-results.json

UT-45 — CSV upload page and dealer dashboard

Field	Specification and result
Requirement / priority	FR-UPLOAD-01-07; FR-DEALER-05-08 / High
Set-up and data	Mocked inventory and listing APIs.
Steps	Choose a category, download its template, upload a CSV; open the dashboard.
Expected result	Template matches the category; the upload opens its report; failures keep the dealer on the page; totals and the stale-stock reminder appear.
Execution	Executed 27 Sep 2026: 8/8 passed
Evidence	apps/frontend/src/portals/dealer/pages/dealerPages.test.tsx; test-evidence/frontend-results.json

UT-46 — Sign-in form

Field	Specification and result
Requirement / priority	FR-USER-04-06 / High

Field	Specification and result
Set-up and data	Mocked sign-in and password reset.
Steps	Sign in as each role; enter wrong details; request a reset.
Expected result	Each role lands on its own portal; errors are explained in plain words; a reset needs an email first.
Execution	Executed 27 Sep 2026: 8/8 passed
Evidence	apps/frontend/src/features/auth/components/LoginForm.test.tsx; test-evidence/frontend-results.json

UT-47 — Notification centre

Field	Specification and result
Requirement / priority	FR-NOTIFY-01-05 / High
Set-up and data	Mocked notification API with read, unread and failed states.
Steps	Open the list; mark one and all read; open a stale-stock reminder.
Expected result	Unread count and times shown; marking read updates the count once; reminders open the right page; an outage never breaks the page.
Execution	Executed 27 Sep 2026: 5/5 passed
Evidence	apps/frontend/src/features/notifications/components/NotificationCenter.test.tsx; test-evidence/frontend-results.json

UT-48 — Marketplace browse, search and filters

Field	Specification and result
Requirement / priority	FR-MARKET-07; FR-SEARCH-01-08; UR-05-08,16 / High
Set-up and data	Mocked buyer API; browse, search, empty and failed results.
Steps	Filter, search, sort and page; open and close the phone filter sheet.
Expected result	Filters and search reach the server; active filters are counted; empty and failed results offer a way forward; Escape closes the sheet and returns focus.
Execution	Executed 27 Sep 2026: 6/6 passed
Evidence	apps/frontend/src/portals/buyer/pages/Marketplace.test.tsx; test-evidence/frontend-results.json

Appendix B – Integration and Contract Test Cases

IT-01 to IT-11 are automated; IT-12 to IT-15 are planned; CT-01 to CT-07 are the contract and consistency checks from Section 3.2.3.

IT-01 — Dealer application persistence

Field	Specification and result
Requirement / priority	FR-DEALER-09 / High
Set-up and data	Empty test database; valid application.
Steps	1. Create an application. 2. Find it by user ID.
Expected result	One pending application with the stored fields.
Execution	Executed 26 Sep 2026: 1/1 passed
Evidence	apps/backend/src/modules/dealers/dealer.repository.test.ts; test-evidence/backend-results.json

IT-02 — Pending applications oldest first

Field	Specification and result
Requirement / priority	FR-ADMIN-03 / High
Set-up and data	Two applications submitted in order.
Steps	1. Create A then B. 2. List pending.
Expected result	Both returned, oldest submission first.
Execution	Executed 26 Sep 2026: 1/1 passed
Evidence	apps/backend/src/modules/admin/admin.repository.test.ts; test-evidence/backend-results.json

IT-03 — Registration uniqueness in MongoDB

Field	Specification and result
Requirement / priority	FR-ETL-19–22; RR-08 / High
Set-up and data	Active, archived and differently formatted plates.
Steps	1. Look up duplicates. 2. Insert a second draft/active listing with the same plate. 3. Relist after archiving.
Expected result	Duplicates found across formats; the database itself rejects a second open listing; archived plates can be relisted.
Execution	Executed 26 Sep 2026: 6/6 passed
Evidence	apps/backend/src/modules/marketplace/listing.repository.test.ts; test-evidence/backend-results.json

IT-04 — Listing update persistence

Field	Specification and result
Requirement / priority	FR-MARKET-05 / High
Set-up and data	Owned listing with a description.
Steps	1. Set description to null. 2. Set a new description.
Expected result	Description removed, then updated, in the stored document.
Execution	Executed 26 Sep 2026: 2/2 passed
Evidence	apps/backend/src/modules/marketplace/listing.service.test.ts; test-evidence/backend-results.json

IT-05 — Audited document access

Field	Specification and result
Requirement / priority	FR-ADMIN-06; PSR-15 / High
Set-up and data	Dealer with documents; deleted documents; invalid index.
Steps	1. Admin opens a document. 2. Open after retention deletion. 3. Open a missing index.
Expected result	Each view audited with who and which file; 410 Gone after deletion without an audit entry; no entry for a missing file.
Execution	Executed 26 Sep 2026: 3/3 passed
Evidence	apps/backend/src/modules/admin/admin.documentAccess.test.ts; test-evidence/backend-results.json

IT-06 — Rate limits with real Redis

Field	Specification and result
Requirement / priority	PSR-13; RR-10 / High
Set-up and data	Limiter with small budget; Redis unavailable; real Redis store.
Steps	1. Exceed the budget. 2. Stop the store. 3. Count only successes. 4. Share one budget via Redis. 5. Exceed upload concurrency.
Expected result	429 in the standard format; requests allowed when Redis is down; one shared budget across instances; 503 with Retry-After beyond the upload limit.
Execution	Executed 26 Sep 2026: 5/5 passed
Evidence	apps/backend/src/shared/middleware/rateLimits.test.ts; test-evidence/backend-results.json

IT-07 — Lease ownership and idempotent writes

Field	Specification and result
Requirement / priority	RR-06-08; FR-ETL-31-32 / High
Set-up and data	Upload job with an expired and a valid lease; repeated batches.
Steps	1. Take over an expired lease and write as the old owner. 2. Try to claim a job with a valid lease. 3. Insert the same CSV row twice. 4. Record the same rejection twice.
Expected result	Only the current owner can write; a valid lease cannot be stolen; one listing per CSV row and one rejection per bad row.
Execution	Executed 26 Sep 2026: 4/4 passed
Evidence	apps/worker/src/repositories/jobSafety.db.test.ts; test-evidence/worker-results.json

IT-08 — Access control across dealers and roles

Field	Specification and result
Requirement / priority	FR-USER-07-12; PSR-08-09; FR-DEALER-10 / High
Set-up and data	Buyer, dealers A and B, admin and a suspended user; listings, uploads, documents and notifications owned by A.
Steps	1. As B, request, edit, delete and retry A's resources by ID. 2. Read drafts and private storage publicly. 3. Call dealer/admin routes without a token and with wrong roles. 4. Use a suspended account's valid token. 5. Approve an applicant with an unverified email.
Expected result	Every foreign access is refused with no change to A's data; drafts and private objects never public; 401/403 as appropriate; suspended accounts blocked; approval refused until the email is verified.
Execution	Executed 26 Sep 2026: 10/10 passed
Evidence	apps/backend/src/test/accessControl.journey.test.ts; test-evidence/backend-results.json

IT-09 — Dealer lifecycle and admin monitoring

Field	Specification and result
Requirement / priority	FR-DEALER-01–13; FR-ADMIN-03–07 / High
Set-up and data	Buyer applying from an existing account; admin; listings of a dealer who is later suspended.
Steps	1. Suspend a dealer and check browse, details and search; reactivate. 2. Edit the dealer profile. 3. Apply as a signed-in buyer; reject; correct and resubmit. 4. Try to apply again once approved. 5. Filter uploads and audit logs by ID and date.
Expected result	Suspended dealers' listings vanish from all public views and return on reactivation; only allowed profile fields change; resubmission keeps history and replaces documents; filters return exactly the matching records.
Execution	Executed 26 Sep 2026: 7/7 passed
Evidence	apps/backend/src/test/dealerLifecycle.journey.test.ts; test-evidence/backend-results.json

IT-10 — Bulk tools, stale stock and discovery

Field	Specification and result
Requirement / priority	FR-DEALER-05–08; Extension (bulk, stale, discovery, small copies) / High
Set-up and data	Two dealers; drafts, active, sold and archived listings; listings of one CSV upload; stale dates; a suspended dealer; photos with small copies.
Steps	1. Bulk publish chosen drafts and all drafts of one upload. 2. Send a rival's IDs. 3. Mark sold, archive, cut prices 10%, delete archived. 4. Send invalid bulk requests and a buyer's request. 5. Check stale counts, ordering and fresh-again actions. 6. Request similar and recommended vehicles. 7. Fetch a small photo copy.
Expected result	Only eligible own listings change and the rest are counted as skipped; prices 6,000,000 → 5,400,000; photos and small copies deleted from storage; invalid requests 400, buyers 403; stale = active and unconfirmed for 60 days (legacy listings by update time); similar and recommended lists are ranked, exclude the seed/viewed and hidden listings; small copy served from thumbs/.
Execution	Executed 26 Sep 2026: 15/15 passed
Evidence	apps/backend/src/test/listingTools.journey.test.ts; test-evidence/backend-results.json

IT-11 — Photo URL updates in MongoDB

Field	Specification and result
Requirement / priority	PSR-13; Extension (small photo copies) / High
Set-up and data	Listing whose photo keys contain dots, as real keys do.
Steps	1. Set the small-copy URL on one photo. 2. Rewrite both photo URLs.
Expected result	Only the listed photos change and all other image fields are kept (regression test for finding F-08).
Execution	Executed 26 Sep 2026: 2/2 passed
Evidence	apps/backend/src/scripts/listingImageMigration.db.test.ts; test-evidence/backend-results.json

IT-12 — Real Firebase identity

Field	Specification and result
Requirement / priority	FR-USER-01–09; PSR-08 / High
Set-up and data	Dedicated test Firebase project or emulator [7]; buyer and dealer test accounts.

Field	Specification and result
Steps	1. Sign in and call /api/v1/auth/me twice. 2. Revoke the refresh token; call again after the cache period. 3. Delete the Firebase user.
Expected result	One local profile per Firebase user; revoked or deleted users lose access within the cache period (2 minutes).
Execution	Planned; not executed
Evidence	Not yet recorded

IT-13 — Real SMTP delivery and failure

Field	Specification and result
Requirement / priority	FR-NOTIFY-06-07 / High
Set-up and data	Test mailbox or local mail sink; failed import and approval events.
Steps	1. Trigger the events. 2. Check received mail. 3. Break SMTP credentials and trigger again.
Expected result	Mail arrives with the right content; with SMTP broken, in-app notifications remain and email status becomes pending then failed after five attempts.
Execution	Planned; not executed
Evidence	Not yet recorded

IT-14 — CSV and ZIP through queue and live worker

Field	Specification and result
Requirement / priority	FR-UPLOAD-04-08; FR-ETL-01-33 / High
Set-up and data	Running backend + worker + Redis + MinIO; CSV with valid, invalid and duplicate rows; ZIP with matching and unknown folders.
Steps	1. Upload the CSV through the API. 2. Wait for completion. 3. Upload the ZIP. 4. Inspect listings, rejected rows, photos and small copies.
Expected result	Counts match the fixture; rejected rows keep their row numbers; photos and small copies attached only to this upload's listings; unknown folder reported.
Execution	Planned; not executed
Evidence	Not yet recorded

IT-15 — Search relevance on a labelled dataset

Field	Specification and result
Requirement / priority	FR-SEARCH-01-20 / High
Set-up and data	Over 300 listings including older exact matches; 20 pre-labelled queries.
Steps	1. Run each query with semantic search on and off. 2. Record the top five.
Expected result	Hard filters always hold; the labelled match is in the top five for at least 18 of 20 queries (proposed target).
Execution	Planned; not executed
Evidence	Not yet recorded

CT-01 — Shared contracts compile into every consumer

Field	Specification and result
Requirement / priority	Maintainability; all FR areas / Medium

Field	Specification and result
Set-up and data	Locked dependencies; Node 24; all four workspaces.
Steps	Run <code>npm run build --workspaces --if-present</code> , which type-checks backend, worker and frontend against packages/shared-contracts.
Expected result	Every workspace builds; no type error at any import of a shared schema or DTO.
Execution	Executed 27 Sep 2026: build exit code 0
Evidence	test-evidence/build-output.txt

CT-02 — Sinhala and Tamil dictionaries agree with English

Field	Specification and result
Requirement / priority	Extension (languages) / Medium
Set-up and data	en.ts, si.ts and ta.ts message files.
Steps	Compare key sets and {placeholder} names of every message (automated in UT-39, first test).
Expected result	Same keys in all three; same placeholders in every translated message.
Execution	Executed 26 Sep 2026 within UT-39: passed
Evidence	apps/frontend/src/shared/i18n/i18n.test.tsx

CT-03 — Queue job IDs agree between API and worker

Field	Specification and result
Requirement / priority	RR-06–08 / Medium
Set-up and data	inventoryBullJobId in shared-contracts; backend publisher; worker reaper.
Steps	Check that the API publishes and the reaper looks up jobs by the same fixed IDs (UT-41 and UT-24).
Expected result	CSV jobs use the upload ID and photo jobs the ID with an -images suffix, on both sides.
Execution	Executed 26 Sep 2026 within UT-24 and UT-41: passed
Evidence	apps/backend/src/modules/inventory/inventory.queue.test.ts; apps/worker/src/jobs/reaper.job.test.ts

CT-04 — Deployment publishes only the website and the API

Field	Specification and result
Requirement / priority	PSR-10–13 / Medium
Set-up and data	compose.yml plus compose.ec2.yml.
Steps	1. Resolve the Compose files and list port bindings. 2. On EC2, run <code>docker compose ps</code> after deployment.
Expected result	3000 and 4173 on all interfaces; Redis 6379 and MinIO 9000/9001 on 127.0.0.1 only.
Execution	Executed 27 Sep 2026: passed locally and on EC2
Evidence	test-evidence/compose-ec2-ports.txt; EC2 docker compose ps output

CT-05 — Frontend accepts the address users open

Field	Specification and result
Requirement / priority	PSR-10; configuration / Medium
Set-up and data	Vite 6.4.3 development server; EC2 public DNS name.
Steps	Request the app with the EC2 Host header, with and without <code>VITE_ALLOWED_HOSTS</code> .

Field	Specification and result
Expected result	Refused without the setting (403), served with it (200); compose.ec2.yml sets it.
Execution	Executed 27 Sep 2026: 403 without, 200 with
Evidence	test-evidence/vite-allowed-hosts.txt

CT-06 — API and worker agree on shared collection shapes

Field	Specification and result
Requirement / priority	RR-02–05 / Medium
Set-up and data	Backend and worker Mongoose models for listings, uploadJobs, notifications and authusers.
Steps	Compare field names, types and indexes declared by both processes for each shared collection.
Expected result	No field written by one process is unknown to, or typed differently by, the other.
Execution	Not automated; checked by code review only
Evidence	Gap recorded as risk R-07

CT-07 — Stored photo addresses match the deployment

Field	Specification and result
Requirement / priority	FR-MARKET-16 / Medium
Set-up and data	Live data; deployed stack.
Steps	Fetch the main photo of every active listing from its stored URL (smoke check).
Expected result	Every stored photo URL points at a host that serves the file.
Execution	Executed 27 Sep 2026: 22 of 23 load on EC2
Evidence	test-evidence/ec2-smoke-results.json; finding F-01

Appendix C – End-to-End Test Cases

E2E-A1 to E2E-A3 are system procedures that were executed. E2E-01 to E2E-15 are browser journeys to be performed on the EC2 deployment; enter each result in manual-results.csv and regenerate this report.

E2E-A1 — Worker killed during a 60,000-row import

Field	Specification and result
Requirement / priority	RR-06–08; FR-ETL-31–32; PSR-05 / High
Set-up and data	Isolated Docker test stack with two worker containers; generated CSV of 60,000 unique rows.
Steps	1. Start the import. 2. Kill the first worker part-way. 3. Let the second worker take over when the 2-minute lease expires. 4. Count listings and distinct source rows.
Expected result	The import completes without manual action; listings = distinct source rows = 60,000.
Execution	Executed Sep 2026: passed — finished by the second worker in 4 min 43 s
Evidence	scripts/drills/kill-worker-mid-import.sh; docs/RESILIENCE.md

E2E-A2 — Read-only smoke of the development stack

Field	Specification and result
Requirement / priority	FR-MARKET-07–11; FR-SEARCH-01–06; PSR-08–09 / High
Set-up and data	Laptop Docker stack rebuilt from the baseline; real Atlas data; GET requests only.
Steps	Run python scripts/smoke/live-stack-smoke.py against the local API and frontend.
Expected result	Health and readiness 200; only active listings public; filters hold; protected routes 401; similar and recommended lists exclude the seed; every active listing photo loads; the app shell is served.
Execution	Executed 2026-09-26: 16 passed, 1 failed, 0 skipped
Evidence	test-evidence/live-smoke-output.txt; live-smoke-results.json

E2E-A3 — Read-only smoke of the EC2 deployment

Field	Specification and result
Requirement / priority	FR-MARKET-07–11; FR-SEARCH-01–06; PSR-01–02; PSR-08–10 / High
Set-up and data	EC2 deployment (Elastic IP 13.207.143.45) rebuilt on Node 24 with compose.ec2.yml; real Atlas data; run from a laptop in Sri Lanka.
Steps	Run the smoke script against http://13.207.143.45:3000 and http://13.207.143.45:4173.
Expected result	As E2E-A2, measured across the internet.
Execution	Executed 2026-09-27: 15 passed, 2 failed; API responses 121–228 ms
Evidence	test-evidence/ec2-smoke-results.json

E2E-01 — Buyer registration, sign-in and session

Field	Specification and result
Requirement / priority	FR-USER-01–06A / High
Set-up and data	Unused test email; existing buyer account.
Steps	1. Open /signup; submit with missing fields and mismatched passwords. 2. Register a valid buyer; open the verification email. 3. Sign out and sign in. 4. Try a wrong password and a duplicate email. 5. Reload /marketplace while signed in.
Expected result	Clear field errors; one account created; verification banner disappears after verifying; wrong password shows “The email address or password is incorrect.”; session survives reload.

Field	Specification and result
Execution	Not executed (manual browser journey)
Evidence	Record screenshots and notes in manual-results.csv

E2E-02 — Dealer application and approval

Field	Specification and result
Requirement / priority	FR-DEALER-09–13; FR-ADMIN-03,06 / High
Set-up and data	Applicant with a PDF registration and ID; admin in a second browser profile.
Steps	1. Apply at /dealer/apply. 2. Check the status page and that /dealer is refused. 3. Admin opens Dealer Approvals from the dashboard link, views both documents and approves. 4. Applicant signs in again.
Expected result	Application appears in Pending (oldest first); document views appear in Audit Logs; applicant reaches the Dealer Dashboard; an approval notification appears.
Execution	Not executed (manual browser journey)
Evidence	Record screenshots and notes in manual-results.csv

E2E-03 — Rejection, correction and resubmission

Field	Specification and result
Requirement / priority	FR-DEALER-11–13 / High
Set-up and data	Pending applicant; admin.
Steps	1. Admin rejects with a reason. 2. Applicant opens the status page and chooses “Correct and resubmit”. 3. Changes a field, attaches new documents and resubmits. 4. Admin reviews again.
Expected result	Reason shown to the applicant; form pre-filled without a password field; resubmission returns to Pending with the earlier rejection shown in its history.
Execution	Not executed (manual browser journey)
Evidence	Record screenshots and notes in manual-results.csv

E2E-04 — Create, edit, publish and photograph a vehicle

Field	Specification and result
Requirement / priority	FR-MARKET-01–05,07–16 / High
Set-up and data	Approved dealer; unique car; two photos; buyer profile.
Steps	1. Add New Vehicle with two photos as a draft. 2. Edit the price. 3. Publish from My Listings. 4. As buyer, open it from the marketplace. 5. Dealer removes one photo; buyer reloads.
Expected result	Draft not visible to the buyer; after publishing, details, price and dealer contacts are correct; removed photo no longer shown; photos load the small copy on a phone-width window (check the Network tab).
Execution	Not executed (manual browser journey)
Evidence	Record screenshots and notes in manual-results.csv

E2E-05 — Recover from a failed photo upload

Field	Specification and result
Requirement / priority	FR-MARKET-12–15; RR-08–09 / High
Set-up and data	New vehicle; block the image request once (DevTools request blocking).

Field	Specification and result
Steps	1. Create the vehicle with two photos while one request is blocked. 2. Read the error. 3. Unblock and use "Retry images". 4. Press Create again once, deliberately.
Expected result	Exactly one listing exists; the missing photo can be added from the edit page; pressing Create again is refused as a duplicate registration, not a second listing.
Execution	Not executed (manual browser journey)
Evidence	Record screenshots and notes in manual-results.csv

E2E-06 — CSV import, corrections and ZIP photos

Field	Specification and result
Requirement / priority	FR-UPLOAD-01-08; FR-ETL-13-22,31-33 / High
Set-up and data	Category template; CSV with valid, invalid and duplicate rows; ZIP with a matching and an unknown folder.
Steps	1. Download the car template. 2. Upload the mixed CSV and watch the status. 3. Read rejected rows. 4. Upload the ZIP. 5. Use "Publish all N" on the upload page. 6. Check the marketplace.
Expected result	Counters match the file; each rejection names the row and reason; unknown folder listed; after Publish all, every valid vehicle is public with its photos; notifications match the result.
Execution	Not executed (manual browser journey)
Evidence	Record screenshots and notes in manual-results.csv

E2E-07 — Buyer search, filters and dealer contact on a phone

Field	Specification and result
Requirement / priority	FR-MARKET-07-11,16; FR-SEARCH-01-20; Extension (mobile) / High
Set-up and data	Seeded catalogue; DevTools device mode at 390 × 844.
Steps	1. Search "automatic SUV under 8 million near Colombo". 2. Open "Filters (n)", add a make, press "Show N vehicles". 3. Try "toyata corola" and a no-match query. 4. Open a vehicle; swipe the photos; tap WhatsApp.
Expected result	Results respect the filters; the sheet shows the active filter count; empty results explain what to do; swipe changes photos; WhatsApp opens with the number in 94... form and the message pre-filled.
Execution	Not executed (manual browser journey)
Evidence	Record screenshots and notes in manual-results.csv

E2E-08 — Similar vehicles, recommendations and compare

Field	Specification and result
Requirement / priority	Extension (discovery, compare) / High
Set-up and data	Catalogue with several makes; fresh browser profile.
Steps	1. Open three vehicles. 2. Return to the marketplace and check "Recommended for you". 3. On a vehicle page, check "Similar vehicles". 4. Add three vehicles to compare; try a fourth. 5. Open "Compare now", remove one, reload, copy the link to another profile.
Expected result	Recommendations appear only after viewing and never include the viewed vehicles; similar vehicles exclude the current one; fourth vehicle refused with a message; best price/year/mileage highlighted; the shared link shows the same comparison.
Execution	Not executed (manual browser journey)
Evidence	Record screenshots and notes in manual-results.csv

E2E-09 — Bulk actions and stale stock

Field	Specification and result
Requirement / priority	FR-DEALER-05–08; Extension (bulk, stale) / High
Set-up and data	Dealer with drafts and at least one active listing whose lastConfirmedAt is older than 60 days (set in the test database).
Steps	1. Open the dashboard and read the stale banner. 2. Review now → Needs attention. 3. Use Still available, Reduce price 10% and Mark sold on different rows. 4. On My Listings, select all drafts and Publish. 5. Archive two and Delete permanently.
Expected result	Counts in the banner and tab match; each action updates the list and the counts; reduced price shown to buyers; skipped listings explained; deleted listings and photos gone.
Execution	Not executed (manual browser journey)
Evidence	Record screenshots and notes in manual-results.csv

E2E-10 — Stale-stock reminder notification

Field	Specification and result
Requirement / priority	Extension (stale stock); FR-NOTIFY-01–03 / High
Set-up and data	Worker running the baseline; dealer with stale listings.
Steps	1. Start the worker. 2. Open the dealer's notification bell. 3. Click the reminder. 4. Restart the worker.
Expected result	One reminder with the correct count; clicking opens the Needs attention list; no second reminder within 7 days after restart.
Execution	Not executed (manual browser journey)
Evidence	Record screenshots and notes in manual-results.csv

E2E-11 — Administrator moderation and suspension

Field	Specification and result
Requirement / priority	FR-ADMIN-01–06,09–10; FR-USER-10–11 / High
Set-up and data	Admin and an active dealer with public listings.
Steps	1. Suspend the dealer. 2. As buyer, search for their vehicle. 3. As the dealer, try to edit a listing. 4. Reactivate. 5. Check Audit Logs filtered by today.
Expected result	Suspended dealer's vehicles disappear from browse, search and details; dealer actions refused; everything returns after reactivation; both actions audited with the admin's name.
Execution	Not executed (manual browser journey)
Evidence	Record screenshots and notes in manual-results.csv

E2E-12 — Responsive layout and keyboard use

Field	Specification and result
Requirement / priority	UR-01–04,09–12,16–18; Extension (mobile) / High
Set-up and data	Widths 360, 390, 768 and 1440 px; keyboard only.
Steps	1. On each width, use the menu in the buyer site and in the dealer and admin portals. 2. Open and close the filter sheet, compare tray and price dialog with Tab/Enter/Escape. 3. View My Listings and Upload Monitoring on a phone width. 4. Tap an input on an iPhone-size width.
Expected result	No page scrolls sideways; menus reachable at every width and close on Escape; focus visible and returned; tables show as labelled cards; inputs do not zoom the page; buttons at least 44 px tall.
Execution	Not executed (manual browser journey)

Field	Specification and result
Evidence	Record screenshots and notes in manual-results.csv

E2E-13 — Sinhala and Tamil

Field	Specification and result
Requirement / priority	Extension (languages) / Medium
Set-up and data	Native Sinhala and Tamil readers if available.
Steps	1. Choose සිංහල in the navbar, browse, search and open a vehicle. 2. Reload. 3. Choose தமிழ் and repeat. 4. Sign in page in each language.
Expected result	All buyer pages and sign-in change language; choice survives reload; text fits on a phone; readers confirm wording is natural (record their comments).
Execution	Not executed (manual browser journey)
Evidence	Record screenshots and notes in manual-results.csv

E2E-14 — Notification centre and email status

Field	Specification and result
Requirement / priority	FR-NOTIFY-01-07 / High
Set-up and data	Dealer and admin; test mailbox.
Steps	1. Trigger a failed import and an approval. 2. Check bell counts. 3. Open and mark read; reload. 4. Check another user's inbox.
Expected result	Right recipient, correct unread count after reload, email status shown; other users see nothing of it.
Execution	Not executed (manual browser journey)
Evidence	Record screenshots and notes in manual-results.csv

E2E-15 — Mobile performance audit

Field	Specification and result
Requirement / priority	PSR-01; Extension (mobile) / Medium
Set-up and data	Production build served locally; Lighthouse in Chrome, mobile preset.
Steps	1. Audit the landing page, marketplace and one vehicle page. 2. Save the reports.
Expected result	Record Performance, Accessibility and Best Practices scores and LCP; compare with the pre-mobile baseline if available.
Execution	Not executed (manual browser journey)
Evidence	Record screenshots and notes in manual-results.csv

Appendix D – Non-functional Cases and Execution Results

NF-01 — 5,000-row import throughput

Field	Specification and result
Requirement / priority	PSR-04–06 / High
Set-up and data	5,000 generated car rows; real pipeline, replica set and MinIO; test laptop.
Steps	Run the benchmark with RUN_BENCHMARKS=1; repeat once.
Expected result	All 5,000 rows imported within 120 s.
Execution	Executed 26 Sep 2026: passed twice — 11.1 s and 11.2 s (≈450 rows/s)
Evidence	apps/worker/src/benchmarks/importThroughput.bench.test.ts; test-evidence/benchmark-output.txt

NF-02 — Response-time percentiles

Field	Specification and result
Requirement / priority	PSR-01–03 / Medium
Set-up and data	EC2 deployment; seeded catalogue; fixed provider settings.
Steps	1. Warm up 2 minutes. 2. Replay browse, filter, search and details requests for 10 minutes with k6. 3. Repeat three times.
Expected result	95th percentile of normal requests ≤ 2 s; structured search ≤ 2 s; semantic search normally ≤ 5 s.
Execution	Partly evidenced: single requests on EC2 took 121–228 ms (E2E-A3); percentile run not executed
Evidence	test-evidence/ec2-smoke-results.json (indicative only)

NF-03 — Concurrent buyers during an import

Field	Specification and result
Requirement / priority	PSR-01; PSR-06–07 / Medium
Set-up and data	EC2 deployment; traffic mix from 3.3.5.
Steps	1. Run 5, 20 and 50 virtual users. 2. Start a 5,000-row import during the 20-user stage. 3. Record latency, errors, CPU, memory and queue depth.
Expected result	Normal-load stage meets PSR-01; under 1% server errors; nothing lost; resources recover.
Execution	Not executed
Evidence	Not yet recorded

NF-04 — Upload limits and unsafe files

Field	Specification and result
Requirement / priority	FR-MARKET-13; PSR-13–14 / Medium
Set-up and data	Pixel bombs, fake images, dangerous PDFs, oversize and over-full ZIPs, parallel uploads.
Steps	Covered by UT-05, UT-09, UT-10, UT-21 and IT-06.
Expected result	Unsafe input refused before it can use much memory; limits answer with clear errors.
Execution	Executed within the referenced cases: passed
Evidence	See UT-05, UT-09, UT-10, UT-21, IT-06

NF-05 — Build and supply-chain checks

Field	Specification and result
Requirement / priority	SRS 2.4; PSR-10–12 / Medium
Set-up and data	Locked dependencies; Node 24; CI workflow.
Steps	1. Build all workspaces. 2. CI runs Gitleaks over the full history, Trivy on each image and npm audit on every push and weekly.
Expected result	Builds succeed; no leaked secret; no unaddressed high or critical vulnerability.
Execution	Executed 27 Sep 2026: build passed; CI scans passed on pull requests #17 and #29 (run #70)
Evidence	test-evidence/build-output.txt; .github/workflows/ci.yml; GitHub Actions runs for PR #17 and PR #29; screenshots/ci-run-70-pr29.png; test-evidence/typecheck-and-audit.txt

NF-06 — Search relevance on labelled queries

Field	Specification and result
Requirement / priority	FR-SEARCH-08–20 / Medium
Set-up and data	Same dataset as IT-15.
Steps	Record precision at five for 20 labelled queries with semantic search on and off.
Expected result	Target: the labelled match in the top five for at least 18 of 20 queries.
Execution	Not executed
Evidence	Not yet recorded

D.1 Results by test file

Case	Test file	Tests	Passed	Failed	Time (s)
UT-01	backend/src/test/db.safety.test.ts	8	8	0	0.0
UT-02	backend/src/config/mongoUri.test.ts	10	10	0	0.0
UT-03	backend/src/modules/admin/admin.validation.test.ts	4	4	0	0.0
UT-04	backend/src/modules/dealers/dealer.validation.test.ts	3	3	0	0.0
UT-05	backend/src/modules/dealers/dealerDocument.content.test.ts	9	9	0	0.0
UT-06	backend/src/modules/inventory/inventory.validation.test.ts	6	6	0	0.0
UT-07	backend/src/modules/inventory/inventory.service.test.ts	8	8	0	0.0
UT-08	backend/src/modules/marketplace/listing.validation.test.ts	12	12	0	0.0
UT-09	backend/src/modules/marketplace/listingImage.service.test.ts	2	2	0	0.0
UT-10	backend/src/shared/utis/imageReencode.test.ts	7	7	0	0.8
UT-11	backend/src/scripts/listingImageMigration.test.ts	6	6	0	0.3
UT-12	backend/src/modules/search/search.queryAnalyzer.test.ts	5	5	0	0.0
UT-13	backend/src/modules/buyers/buyer.similarity.test.ts	5	5	0	0.0
UT-14	backend/src/shared/utis/pagination.test.ts	2	2	0	0.0
UT-15	backend/src/shared/middleware/verifyFirebaseToken.test.ts	4	4	0	0.0
UT-16	worker/src/pipeline/extract.test.ts	2	2	0	0.0
UT-17	worker/src/pipeline/normalize.test.ts	6	6	0	0.0
UT-18	worker/src/pipeline/transform.test.ts	3	3	0	0.0
UT-19	worker/src/pipeline/validate.test.ts	17	17	0	0.0
UT-20	worker/src/services/uploadJob.service.test.ts	9	9	0	0.1

Case	Test file	Tests	Passed	Failed	Time (s)
UT-21	worker/src/services/imageProcessing.service.test.ts	17	17	0	1.4
UT-22	worker/src/services/jobLease.test.ts	4	4	0	0.0
UT-23	worker/src/services/transientError.test.ts	12	12	0	0.0
UT-24	worker/src/jobs/reaper.job.test.ts	6	6	0	0.0
UT-25	worker/src/jobs/emailOutbox.job.test.ts	7	7	0	0.0
UT-26	worker/src/jobs/documentRetention.job.test.ts	3	3	0	0.0
UT-27	worker/src/jobs/staleListingReminder.job.test.ts	3	3	0	0.0
UT-28	frontend/src/features/auth/components/RoleGuard.test.tsx	4	4	0	0.1
UT-29	frontend/src/features/auth/components/EmailVerificationBanner.test.tsx	4	4	0	0.4
UT-30	frontend/src/features/auth/context/AuthProvider.test.tsx	2	2	0	0.2
UT-31	frontend/src/features/auth/pages/RegisterPage.test.tsx	3	3	0	0.5
UT-32	frontend/src/portals/dealer/pages/DealerProfile.test.tsx	3	3	0	0.6
UT-33	frontend/src/portals/dealer/pages/UploadDetails.test.tsx	5	5	0	0.6
UT-34	frontend/src/portals/dealer/pages/ListingManager.test.tsx	5	5	0	0.8
UT-35	frontend/src/portals/admin/pages/adminPages.test.tsx	6	6	0	0.3
UT-36	frontend/src/portals/buyer/pages/VehicleDetails.test.tsx	4	4	0	0.5
UT-37	frontend/src/features/compare/compare.test.tsx	4	4	0	0.7
UT-38	frontend/src/shared/components/mobileLayout.test.tsx	3	3	0	0.5
UT-39	frontend/src/shared/i18n/i18n.test.tsx	4	4	0	0.4
UT-40	frontend/src/shared/utils/phone.test.ts	3	3	0	0.0
UT-41	backend/src/modules/inventory/inventory.queue.test.ts	4	4	0	0.0
UT-42	frontend/src/shared/services/apiServices.test.ts	13	13	0	0.0
UT-43	frontend/src/portals/admin/pages/adminManagement.test.tsx	9	9	0	2.4
UT-44	frontend/src/portals/dealer/pages/ListingForm.test.tsx	8	8	0	7.8
UT-45	frontend/src/portals/dealer/pages/dealerPages.test.tsx	8	8	0	1.0
UT-46	frontend/src/features/auth/components/LoginForm.test.tsx	8	8	0	4.8
UT-47	frontend/src/features/notifications/components/NotificationCenter.test.tsx	5	5	0	0.8
UT-48	frontend/src/portals/buyer/pages/Marketplace.test.tsx	6	6	0	2.9
IT-01	backend/src/modules/dealers/dealer.repository.test.ts	1	1	0	0.1
IT-02	backend/src/modules/admin/admin.repository.test.ts	1	1	0	0.1
IT-03	backend/src/modules/marketplace/listing.repository.test.ts	6	6	0	0.1
IT-04	backend/src/modules/marketplace/listing.service.test.ts	2	2	0	0.2
IT-05	backend/src/modules/admin/admin.documentAccess.test.ts	3	3	0	0.1
IT-06	backend/src/shared/middleware/rateLimits.test.ts	5	5	0	0.2
IT-07	worker/src/repositories/jobSafety.db.test.ts	4	4	0	0.1
IT-08	backend/src/test/accessControl.journey.test.ts	10	10	0	0.9
IT-09	backend/src/test/dealerLifecycle.journey.test.ts	7	7	0	0.6
IT-10	backend/src/test/listingTools.journey.test.ts	15	15	0	1.0
IT-11	backend/src/scripts/listingImageMigration.db.test.ts	2	2	0	0.1

Appendix E – Evidence Index

File (docs/test-evidence/)	What it shows
backend-results.json, worker-results.json, frontend-results.json	Raw Vitest results for every automated test.
automated-test-register.csv	Every automated test with its case group, status and duration.
benchmark-output.txt	Two consecutive 5,000-row imports (NF-01).
live-smoke-output.txt, live-smoke-results.json	Development-stack smoke run (E2E-A2).
ec2-smoke-results.json	EC2 smoke run with per-check timings (E2E-A3, CT-07).
compose-ec2-ports.txt	Resolved port bindings of the EC2 configuration (CT-04).
vite-allowed-hosts.txt	Host-name check of the frontend server (CT-05).
build-output.txt	Build of all four workspaces (CT-01, NF-05).
coverage-backend.json, coverage-worker.json, coverage-frontend.json	Code coverage per file and in total (Section 5.2).
typecheck-and-audit.txt	Type-check of all three workspaces and npm audit of the production dependencies (Section 5.1.3, NF-05, F-18).
run-metadata.json	Commit, runtimes and commands for this report.
manual-results.csv	Results of manual cases, filled in by testers.
screenshots/	Screenshots of the running system (Appendix H) and of the testing tools: test logs, type-check and audit, Network panel (Section 5.1).
screenshots/lighthouse-*.png	Lighthouse results before and after the production build (Section 5.1.4); saved reports for the remaining E2E-15 pages to be added.

Appendix F – Defect Log

ID	Finding	Severity	Priority	Found by	Status and proof
F-01	Photo URLs are stored with the host that was current at upload time, and the files exist only on that machine. On EC2, 22 of 23 active listings' photos load; one listing's three photos exist only on a laptop.	High	Before demo	E2E-A2, E2E-A3, CT-07	Partly fixed: EC2 restored and laptop photos migrated; re-upload the remaining listing's photos and rerun E2E-A3.
F-02	The 5,000-row benchmark cleared collections before its models loaded, so a second run rejected every row as a duplicate.	Medium	Before submission	NF-01 rerun	Closed: passed twice in a row after the fix.
F-03	Photos uploaded before small copies existed had no 800 px version.	Low	Next iteration	Design review	Partly fixed: 91 laptop photos migrated; run the migration on EC2.
F-04	SRS FR-NOTIFY-06 expects completion e-mails; a clean import only notifies in the app.	Medium	Before submission	Requirement review	Open: team decision needed, then IT-13.
F-05	Sinhala and Tamil text not reviewed by native speakers.	Medium	Before submission	Review of UT-39 scope	Open: E2E-13.
F-06	After a failed photo upload, pressing Create again could attempt a second listing.	Low	Before submission	Code review (v1.0)	Fixed by the version4 merge (a retry updates the saved listing); confirm in E2E-05.
F-07	Dealer profile editing and resubmission after rejection were missing.	High	Before submission	Requirement review (v1.0)	Closed: IT-09, UT-31 and UT-32 pass.
F-08	The photo migration crashed on its first database write because photo keys contain dots.	High	Before demo	First real migration run	Closed: IT-11 on real MongoDB; migration completed with no failures.
F-09	Attaching photos to the same upload a second time did nothing, because the queue kept the finished job's ID.	High	Before demo	Teammate report (version4)	Closed: UT-41.
F-10	Merging version4 produced duplicate code that would not compile and a review-history change that would make approvals fail.	High	Before demo	Merge simulation	Closed: this branch's versions kept, unique fixes ported, 348/348 tests pass.
F-11	A backend test loaded the real Firebase setup and failed only in CI, where credentials are placeholders.	Medium	Before submission	CI on PR #17	Closed: Firebase replaced in the test; the suite passes with placeholder credentials.
F-12	The frontend server refused the EC2 host name; it had only worked through a manual edit on the server.	Medium	Before demo	Deployment review; CT-05	Closed: allow-list set in compose.ec2.yml; 200 with, 403 without.
F-13	Building three images at once on EC2 failed with ETXTBSY under memory pressure.	Medium	Before demo	EC2 deployment	Mitigated: sequential builds with swap; risk R-04 remains.
F-14	The previous EC2 deployment published Redis (no password) and MinIO on all interfaces; only the security group blocked them.	High	Before demo	Deployment review; CT-04	Closed: the override binds both to 127.0.0.1; verified with docker compose ps.
F-15	The EC2 deployment is reached over plain HTTP on ports 4173 and 3000, so the phone browser marks it as not secure and sign-in tokens cross the network unencrypted. PSR-10 requires HTTPS in production.	High	Before demo	EC2 screenshots (Figures 12 and 18)	Open: serve the site and API through a TLS reverse proxy on port 443, then close 3000 and 4173 in the security group.

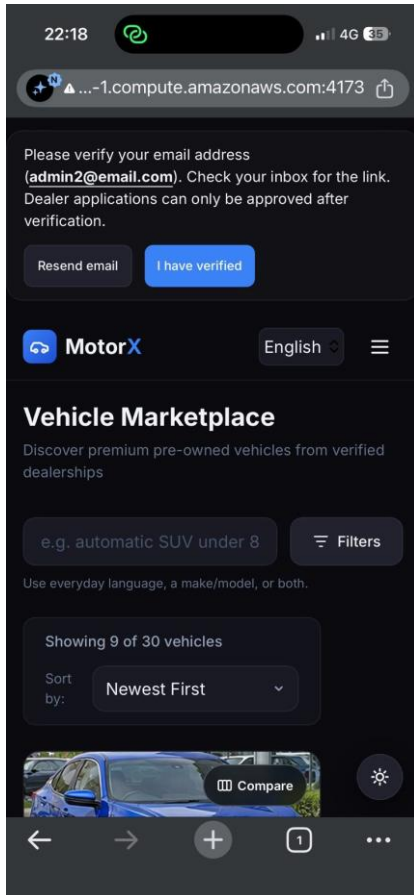
ID	Finding	Severity	Priority	Found by	Status and proof
F-16	The dealer dashboard's upload list prints the raw job status, such as "completedWithErrors", instead of a readable, translated label (UR-02, UR-11).	Low	Next iteration	Screenshot review (Figure 16)	Open: map each status to a readable label, as the admin dashboard already does, and add it to the Sinhala and Tamil texts.
F-17	On 14 Aug 2026 GitHub allowed a pull request into main to be merged while its CI check was still running, so passing CI was not a required status check; a failing build could have been merged.	Low	Before submission	Screenshot review (Figure 20)	Open: in the main branch protection rule, require the CI checks to pass before merging, then confirm the merge button is blocked while a check runs.
F-18	npm audit of the production dependencies reports 12 moderate advisories and no high or critical ones. Two are on code paths MotorX uses: the csv-parse columns option, which reads the header row of dealer-uploaded CSV files, and qs, which Express uses to parse every query string, including public search.	Medium	Before submission	npm audit, 28 Sep 2026 (Section 5.1.3)	Open: run npm audit fix (qs) and rerun the backend tests; upgrade csv-parse to 7.x or reject reserved header names with a worker test; upgrade firebase-admin in a later cycle.

Appendix G – Traceability Matrix

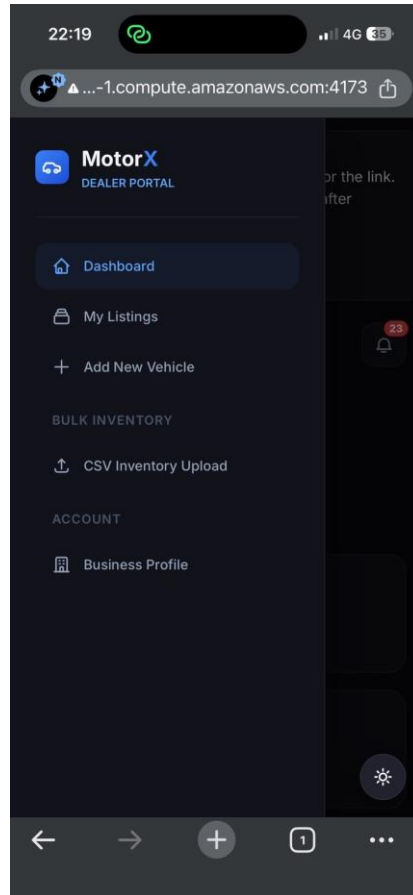
Requirement group	Cases	Evidence status
FR-USER-01–12; PSR-08–09	UT-46, UT-15, UT-28, UT-29, UT-30, IT-06, IT-08, E2E-01, E2E-11, IT-12	7 of 10 executed; all passed
FR-DEALER-01–13	UT-45, UT-04, UT-05, UT-26, UT-31, UT-32, IT-01, IT-05, IT-09, E2E-02, E2E-03	9 of 11 executed; all passed
FR-MARKET-01–16	UT-42, UT-44, UT-48, UT-08, UT-09, UT-10, UT-11, UT-36, IT-03, IT-04, IT-11, CT-07, E2E-04, E2E-05, E2E-07	12 of 15 executed; 1 failed
FR-UPLOAD-01–08	UT-45, UT-06, UT-07, UT-33, UT-41, E2E-06, IT-14	5 of 7 executed; all passed
FR-ETL-01–33	UT-16, UT-17, UT-18, UT-19, UT-20, UT-21, UT-22, UT-23, UT-24, IT-03, IT-07, E2E-A1, E2E-06	12 of 13 executed; all passed
FR-SEARCH-01–20	UT-48, UT-12, UT-14, IT-10, E2E-A2, E2E-A3, E2E-07, IT-15, NF-06	6 of 9 executed; 2 failed
FR-NOTIFY-01–07	UT-47, UT-25, UT-27, E2E-10, E2E-14, IT-13	3 of 6 executed; all passed
FR-ADMIN-01–10	UT-43, UT-03, UT-35, IT-02, IT-05, IT-09, E2E-02, E2E-11	6 of 8 executed; all passed
UR-01–18	UT-48, UT-28, UT-36, UT-38, UT-40, E2E-07, E2E-12	5 of 7 executed; all passed
PSR-01–07	NF-01, NF-02, NF-03, E2E-A3	2 of 4 executed; 1 failed
PSR-08–16	UT-01, UT-02, UT-05, UT-09, UT-10, UT-15, IT-05, IT-06, IT-08, CT-04, CT-05, NF-04, NF-05	All executed and passed
RR-01–10	UT-07, UT-20, UT-22, UT-23, UT-24, UT-25, IT-07, CT-03, E2E-A1	All executed and passed
Extensions — mobile, languages, discovery, comparison, stock tools, small photo copies	UT-11, UT-13, UT-27, UT-33, UT-34, UT-36, UT-37, UT-38, UT-39, UT-40, IT-10, CT-02, E2E-08, E2E-09, E2E-12, E2E-13, E2E-15	12 of 17 executed; all passed

Appendix H – Screenshot Evidence

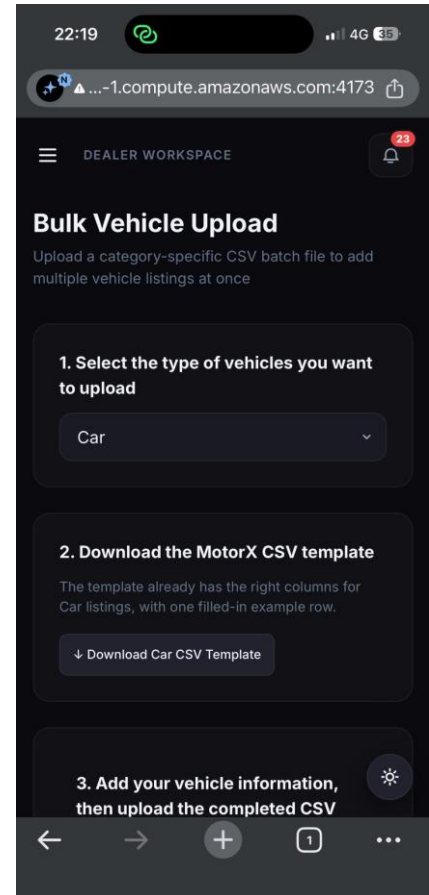
The screenshots below were taken by the team while using the running system, all on 27 September 2026 except Figure 20 (14 August 2026): Figures 12 and 18 on the EC2 deployment (a phone on mobile data, and the AWS console), Figures 19 and 20 on GitHub, and the others in a desktop browser. They show that the pages exist and render with real data; they do not replace the step-by-step journeys in Appendix C, whose results stay as recorded until a tester completes them. Test accounts appear in some images; no customer data is shown.



(a) Marketplace



(b) Dealer portal menu



(c) Bulk vehicle upload

Figure 12: MotorX on the EC2 deployment, opened on a phone over 4G through the public host name (port 4173)

Figure 12 confirms that the deployment answers on its public DNS name from outside the AWS network (CT-05, E2E-A3), and that the buyer marketplace, the dealer navigation and the bulk-upload page are laid out for a phone screen (UR-16–17; supporting E2E-07 and E2E-12). The warning sign in the address bar is the browser reporting a plain-HTTP connection, recorded as F-15.

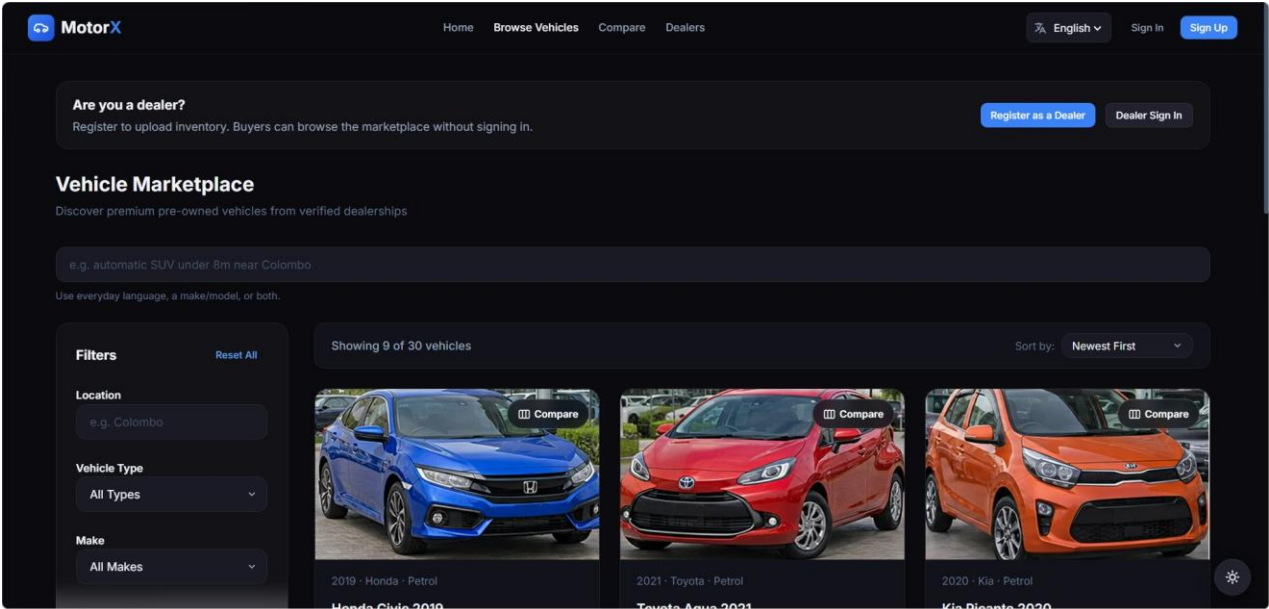


Figure 13: Buyer marketplace on a desktop browser: natural-language search box, filters, sorting, result count and Compare buttons

Figure 13 shows FR-MARKET-07 and FR-SEARCH-01-06 on screen: 30 active vehicles with their photos, the search box that accepts everyday language, the filter panel, “Newest First” sorting, and the Compare toggle on each card (E2E-07, E2E-08).

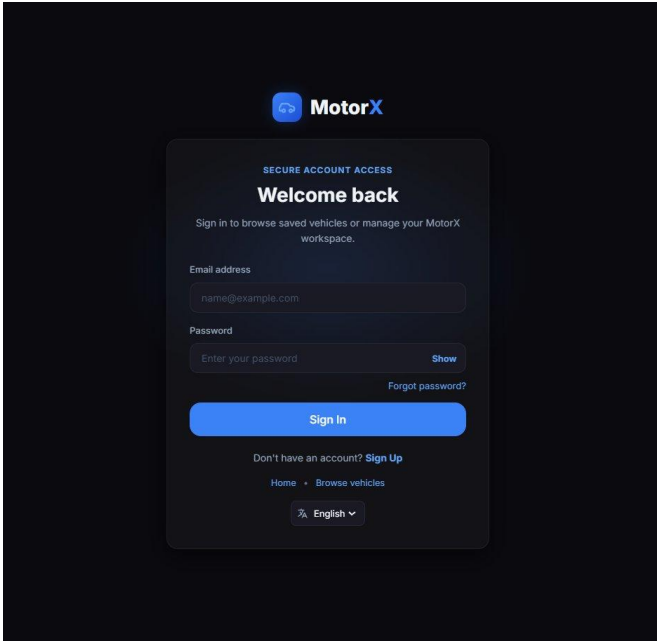


Figure 14: Sign-in page backed by Firebase Authentication, with password reset and the language selector

MotoX

Dealer Application

Submit your dealership details for admin review before your dealer account is activated.

Account Information

Dealer Representative Name

Email Address

Phone Number

Password

Show

Confirm Password

Dealership Information

Dealership / Business Name

Business Registration Number

Business Address

City / District

Province

Business Contact Number

Business Email

Dealership Website or Social Page (optional)

Dealer Details

Type of Dealership

Bath

Main Vehicle Brands Handled (optional)

Short Business Description

Approximate Number of Vehicles Normally Available (optional)

Verification Information

Verification Documents

Business registration
PDF - JPG or PNG max 10 MB

Choose File No file chosen

Representative identity proof
PDF - JPG or PNG max 10 MB

Choose File No file chosen

Additional verification (Optional)
PDF - JPG or PNG max 10 MB

Choose File No file chosen

Back To Login

Submit Dealer Application

Figure 15: Dealer application form: representative, dealership and dealer details, and the three verification-document uploads

Figures 14 and 15 show the entry points for E2E-01 and E2E-02: sign-in with password reset (FR-USER-04-05) and the dealer application with its required fields and document uploads (FR-DEALER-09; PDF, JPG or PNG up to 10 MB, checked by UT-04, UT-05 and IT-01).



MotorX
DEALER PORTAL

Dashboard
My Listings
Add New Vehicle

BULK INVENTORY
CSV Inventory Upload

ACCOUNT
Business Profile

dealer2

Authorized Dealer



Please verify your email address (dealer2@gmail.com). Check your inbox for the link. Dealer applications can only be approved after verification.
Verification email sent.

Resend email
I have verified

DEALER WORKSPACE

Dealer Dashboard
Your inventory overview

Upload CSV
Add Listing

Total Listings

67

Active

22

Draft

29

Sold

0

Archived

16

Recent Inventory

View all (67)

VEHICLE	PRICE	STATUS	MILEAGE	LOCATION
2021 Suzuki Access 125	LKR 540,000	Active	14,000 km	Colombo
2022 Yamaha R15 V3	LKR 950,000	Active	9,000 km	Colombo
2023 Tvs Ntorq 125	LKR 550,000	Active	11,000 km	Colombo
2018 Suzuki G125	LKR 550,000	Active	11,000 km	Colombo
2022 Honda Xr150i	LKR 550,000	Active	11,000 km	Colombo

Upload Activity

View all

motorx-motorcycle-test-listings-new.csv

Completed

motorx-car-test-listings-new.csv

Completed

motorx-motorcycle-test-listings (1).csv

CompletedWithErrors

motorx-car-test-listings (1).csv

Completed

Figure 16: Dealer dashboard: listing totals by status, recent inventory and recent CSV uploads with their final status

Figure 16 shows FR-DEALER-05-07 and FR-ETL-31: 67 listings split into active, draft, sold and archived, and four CSV imports ending as “completed” or “completedWithErrors”. The raw status wording is finding F-16. The banner above the dashboard asks the dealer to verify their e-mail address before an application can be approved.

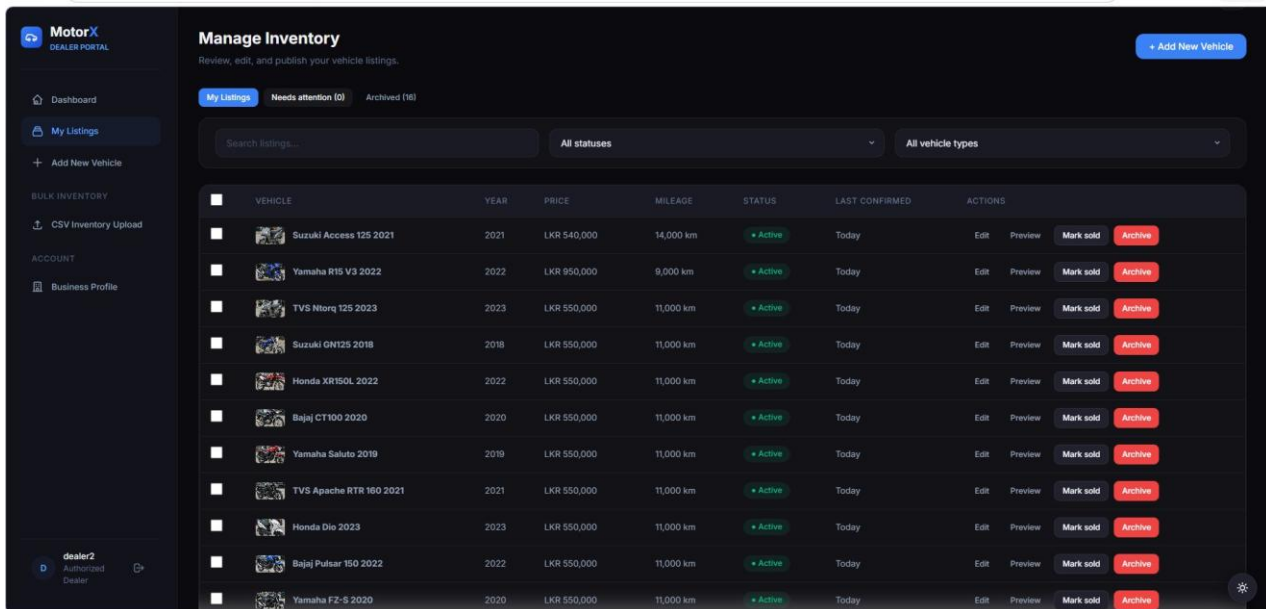


Figure 17: Manage Inventory: tabs for listings needing attention and archived stock, a “Last confirmed” column, row selection and one-click Mark sold / Archive

Figure 17 shows the stock tools added in this cycle (Extension: bulk actions and stale stock; E2E-09): the “Needs attention” tab that collects stale listings, the date each listing was last confirmed, check boxes for bulk actions, and the per-row Mark sold and Archive buttons (UT-27, UT-34, IT-10).

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
-	sgr-0ffad2c70592b519b	IPv4	HTTP	TCP	80	0.0.0.0/0	-
-	sgr-0a546a76fd84c3c0c	IPv4	SSH	TCP	22	175.157.87.141/32	-
-	sgr-0bf4bd0aab248dbcb	IPv4	HTTPS	TCP	443	0.0.0.0/0	-
-	sgr-0551fef21e4c314a9	IPv4	Custom TCP	TCP	3000	0.0.0.0/0	Temporary MotorX bac...
-	sgr-0a45748264fc07348	IPv4	Custom TCP	TCP	4173	0.0.0.0/0	Temporary MotorX fro...

Figure 18: Inbound rules of the EC2 security group motorx-staging-sg (account details cropped out)

Figure 18 is the network-level half of CT-04. SSH is open to a single address, and web traffic to ports 80 and 443. Ports 3000 (API) and 4173 (website) are open to everyone and marked temporary; they stay open until F-15 is fixed, after which only 443 (and 80, redirecting to it) should remain. Redis (6379) and MinIO (9000/9001) have no rule, which matches the 127.0.0.1 binding checked in compose-ec2-ports.txt (F-14).

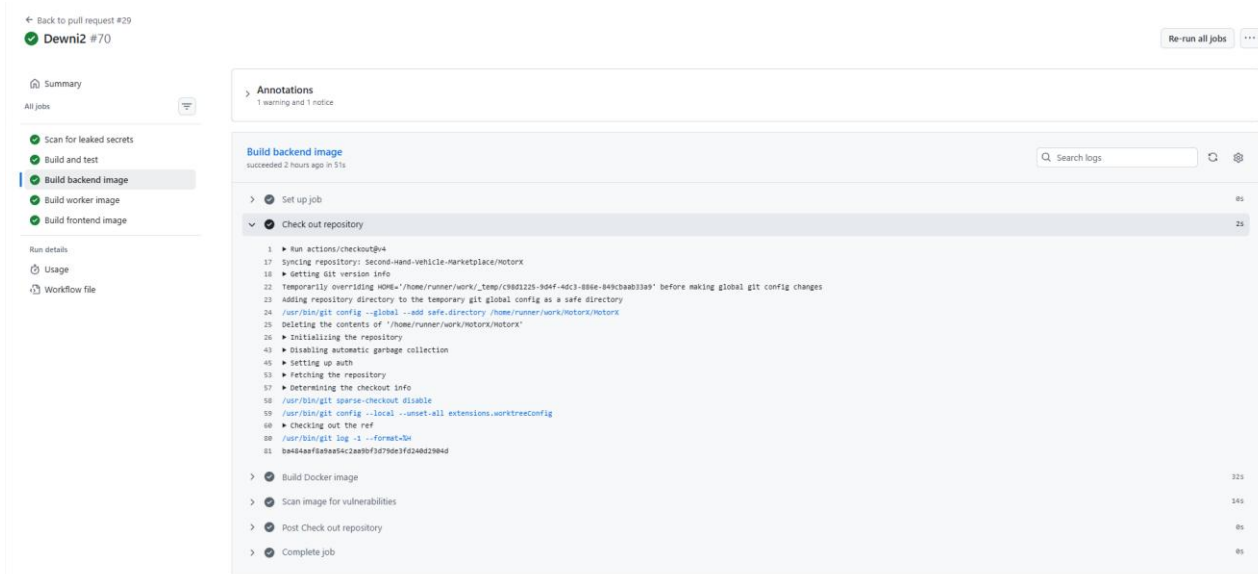


Figure 19: GitHub Actions run #70 for pull request #29 (branch Dewni2): all five CI jobs passed

Figure 19 is the CI evidence for NF-05 and for the automated suites. Every job of the MotorX workflow passed on this pull

request: the scan of the full history for leaked secrets, the build-and-test job that runs the automated tests, and the backend, worker and frontend image builds, each followed by a vulnerability scan of the built image. The checkout step shows the repository and the exact commit tested (ba484aaf). The run also carries one warning and one notice annotation, not shown here.

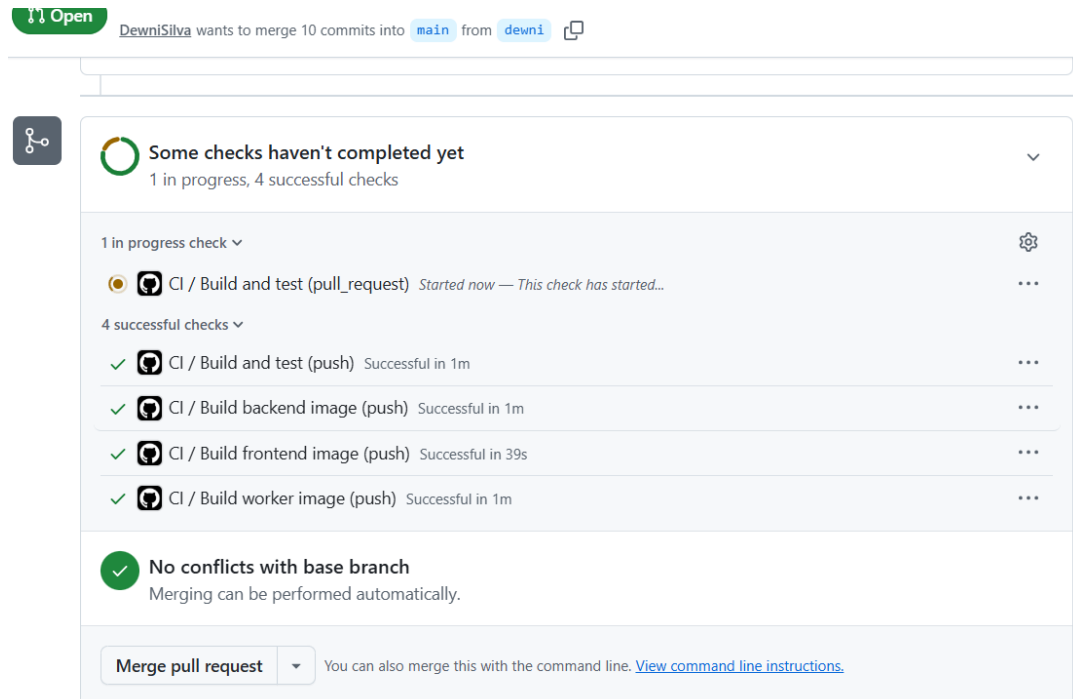


Figure 20: Checks on an earlier pull request from branch dewni into main (14 Aug 2026), taken while the pull-request build was still running

Figure 20 shows that CI runs on both events: the four checks triggered by the push (build and test, and the backend, frontend and worker images) had passed, and the same build-and-test job triggered by the pull request had just started. It also shows that GitHub offered “Merge pull request” before that check finished, so at the time passing CI was not a required condition for merging into main (F-17).